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(54) **ACCURATE REDUCED GATE-DRIVE CURRENT LIMITER**

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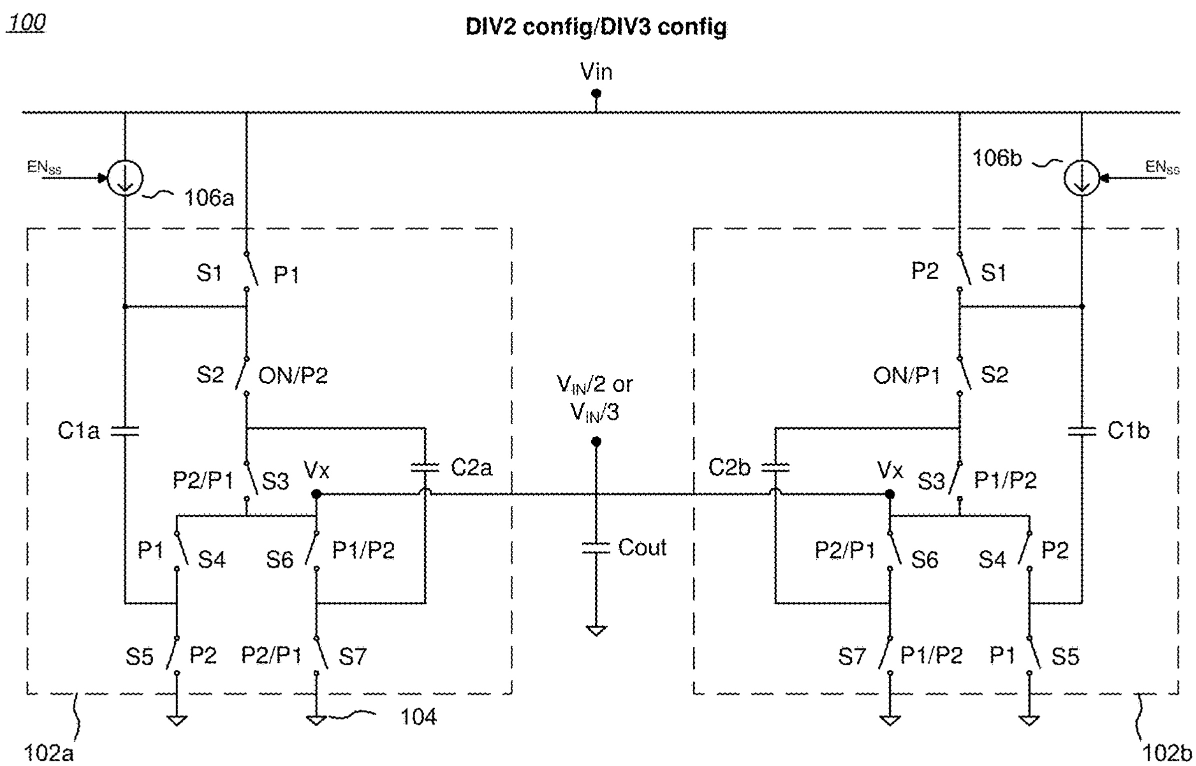
(57) **ABSTRACT**

Circuits and methods that limit current through power FETs of power converter to reduce damaging current in-rush events, independent of switching frequency, device mismatches, and PVT variations. Embodiments utilize a closed-loop feedback circuit and/or a calibrated compensation circuit to regulate, substantially independent of frequency, the control voltage V_{GATE} applied to a power FET gate. In a reduced gate-drive mode, connecting a feedback or compensation circuit to the gate of an LDO source-follower FET allows the gate voltage to be regulated to control the LDO output voltage to a final inverter coupled to the gate of a power FET so that V_{GATE} is adjusted to provide a reduced gate-drive to the power FET; connecting to the output of the LDO allows the LDO output voltage to the final inverter to be directly regulated to adjust V_{GATE}; connecting to the gate of the power FET allows V_{GATE} to be directly set.



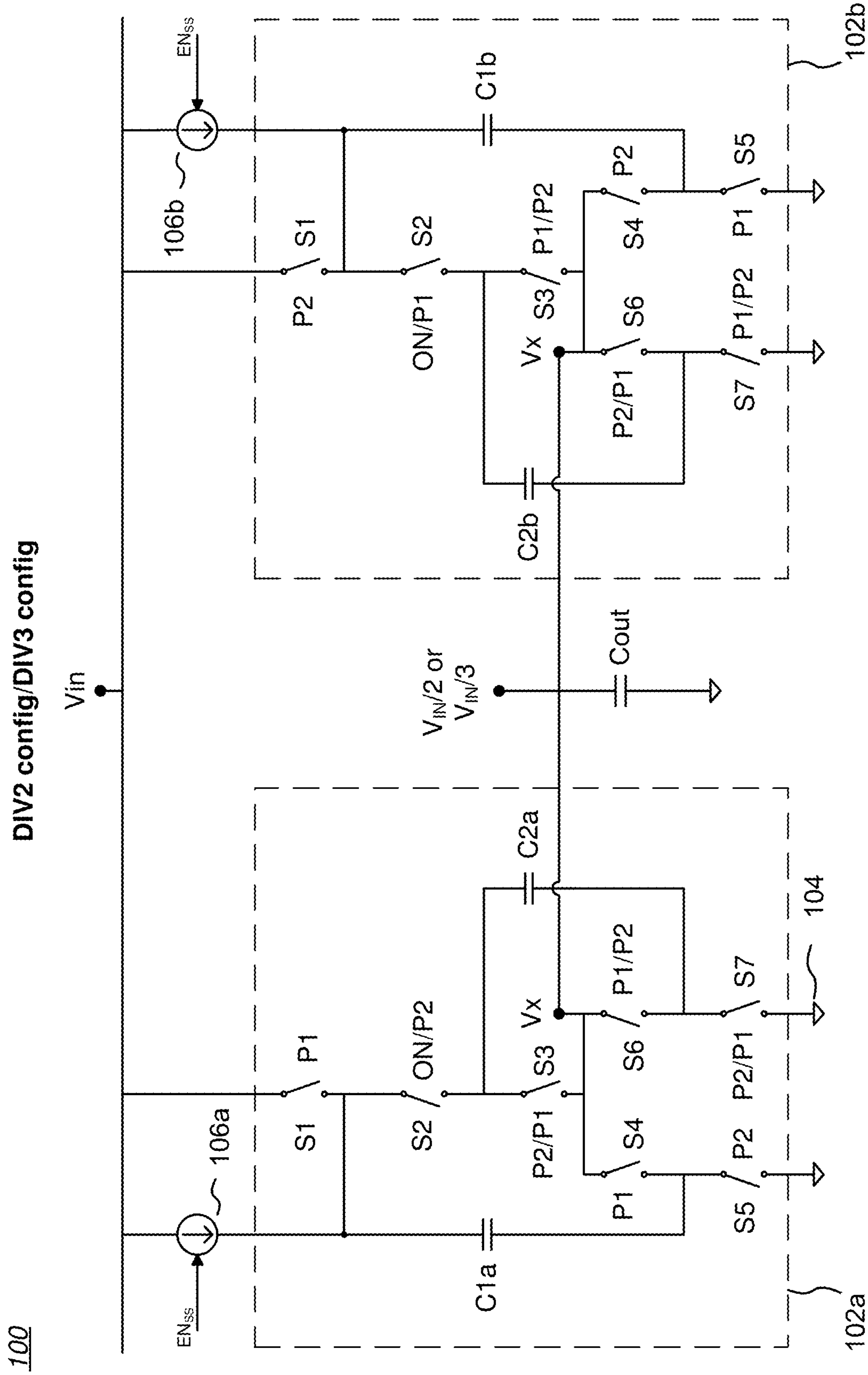

DIV2 config/DIV3 config
100
Vin
EN_{SS}
106a
106b
102a
102b
104
S1
S2
S3
S4
S5
S6
S7
P1
P2
ON/P2
ON/P1
P2/P1
P1/P2
Vx
C1a
C2a
C1b
C2b
$V_{IN}/2$ or
$V_{IN}/3$
Cout

FIG. 1

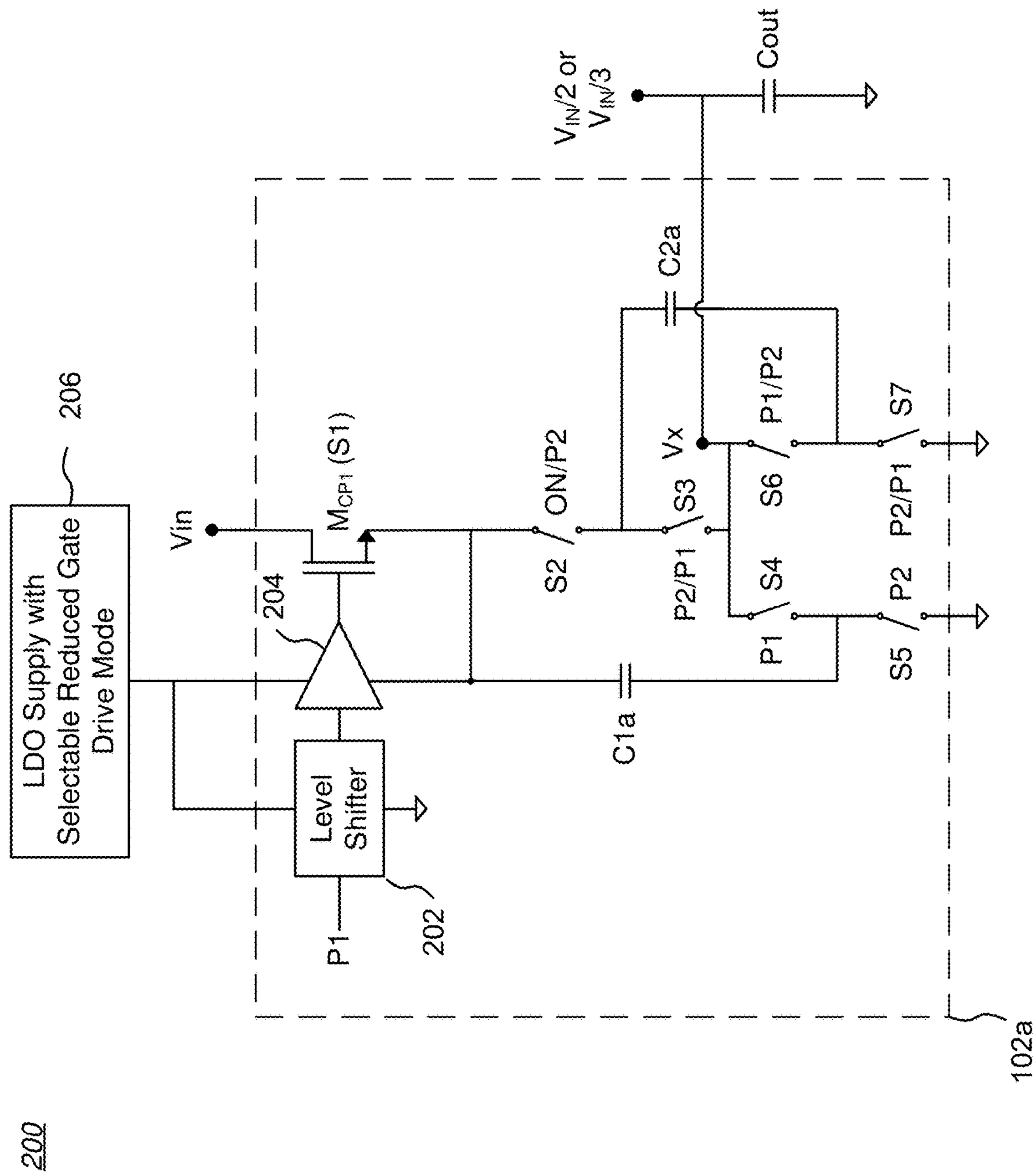

200
LDO Supply with Selectable Reduced Gate Drive Mode
206
Vin
M_{CP1} (S1)
204
P1
Level Shifter
202
S2
ON/P2
S3
P2/P1
S4
P1
S5
P2
S6
P2/P1
S7
P1/P2
Vx
C1a
C2a
$V_{IN}/2$ or $V_{IN}/3$
Cout
102a

FIG. 2

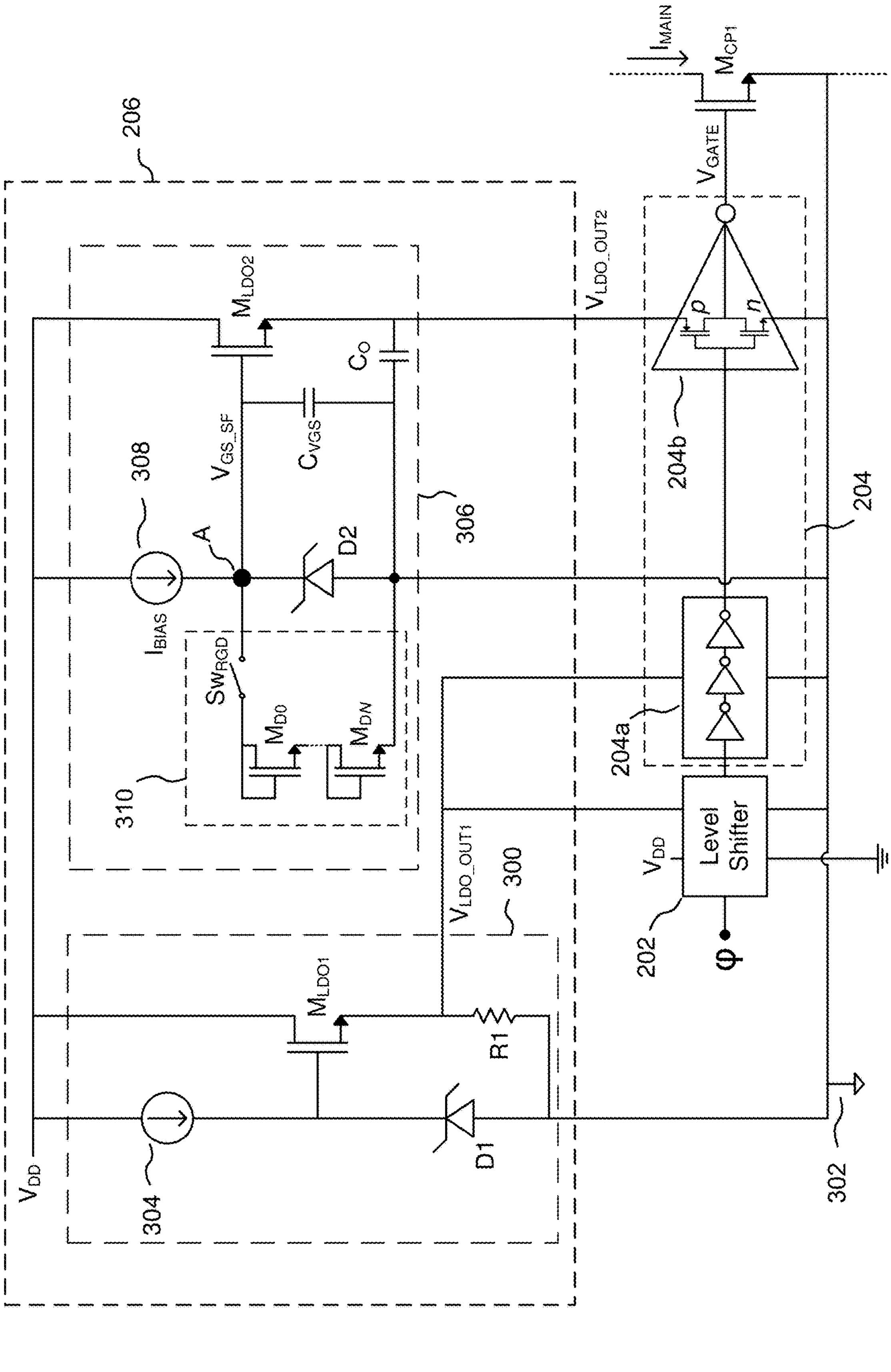

206
M_{LDO2}
V_{LDO_OUT2}
I_{MAIN}
M_{CP1}
V_{GATE}
p
n
C_O
C_{VGS}
V_{GS_SF}
308
A
D2
306
204b
204
I_{BIAS}
Sw_{RGD}
M_{D0}
M_{DN}
310
204a
Level Shifter
V_{DD}
V_{LDO_OUT1}
300
202
Φ
M_{LDO1}
R1
D1
V_{DD}
304
302

FIG. 3

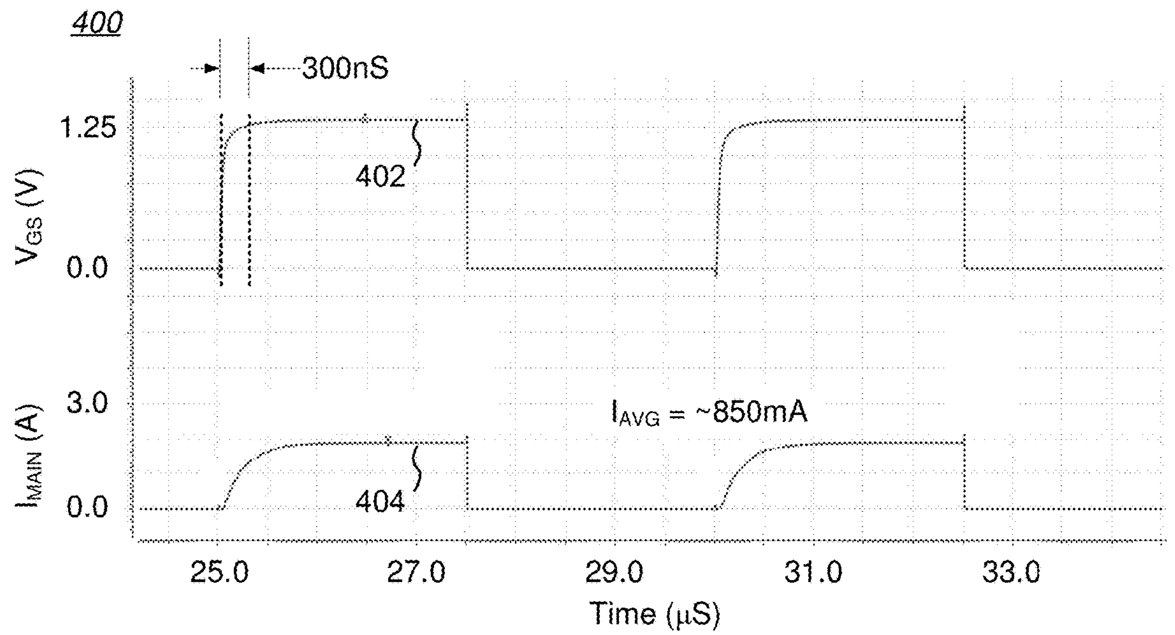


FIG. 4A
(200 kHz)

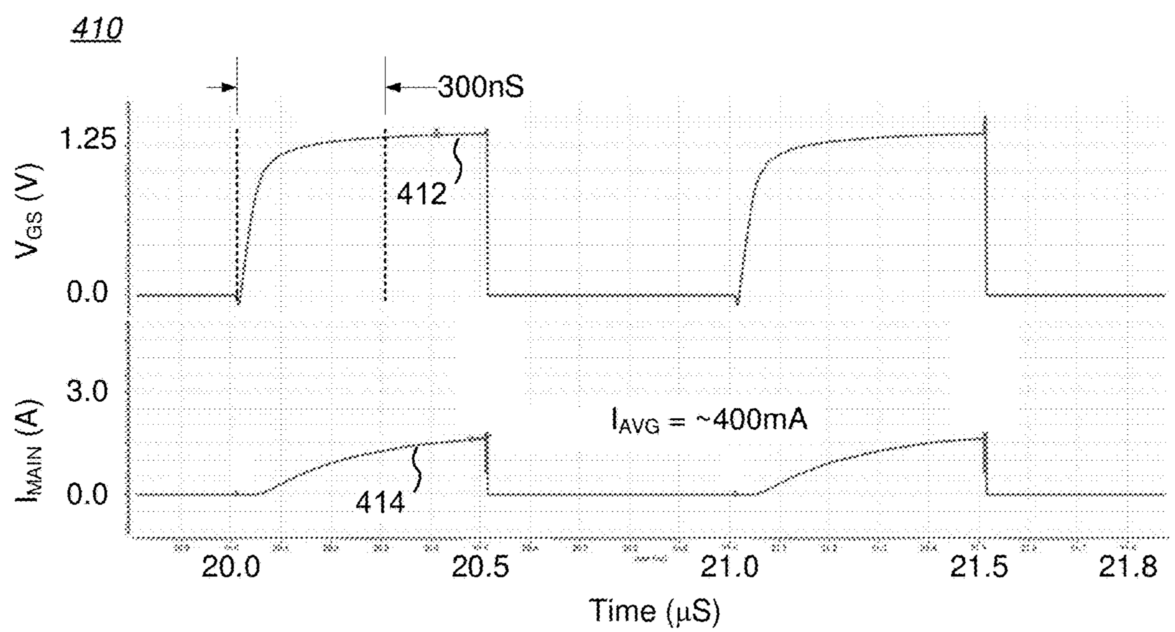


FIG. 4B
(1 MHz)

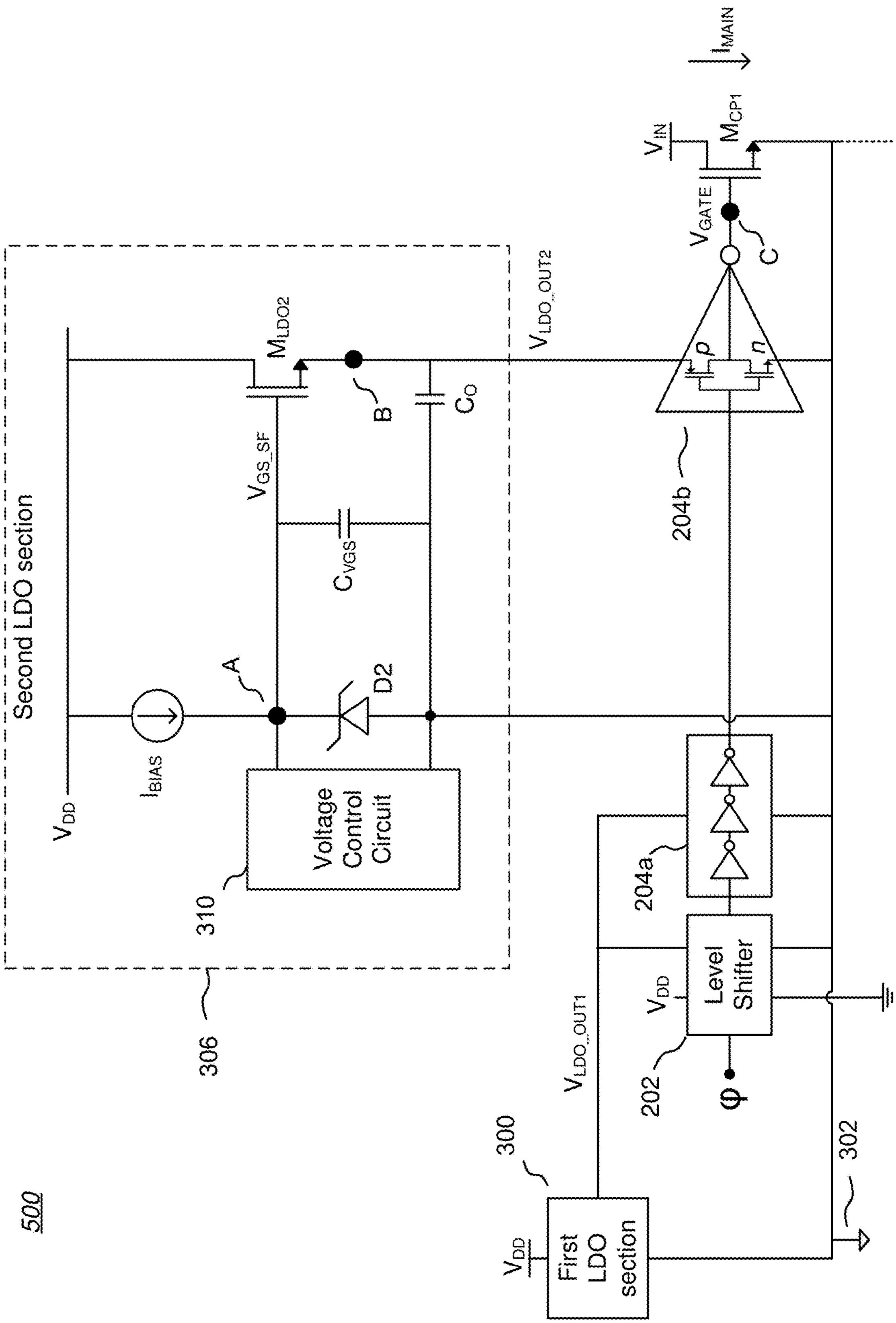

500
Second LDO section
V_{DD}
I_{BIAS}
A
D2
Voltage Control Circuit
310
306
V_{GS_SF}
C_{VGS}
M_{LDO2}
B
C_O
V_{LDO_OUT2}
p
n
204b
V_{GATE}
C
V_{IN}
M_{CP1}
I_{MAIN}
204a
V_{DD}
Level Shifter
202
φ
V_{LDO_OUT1}
300
V_{DD}
First LDO section
302

FIG. 5

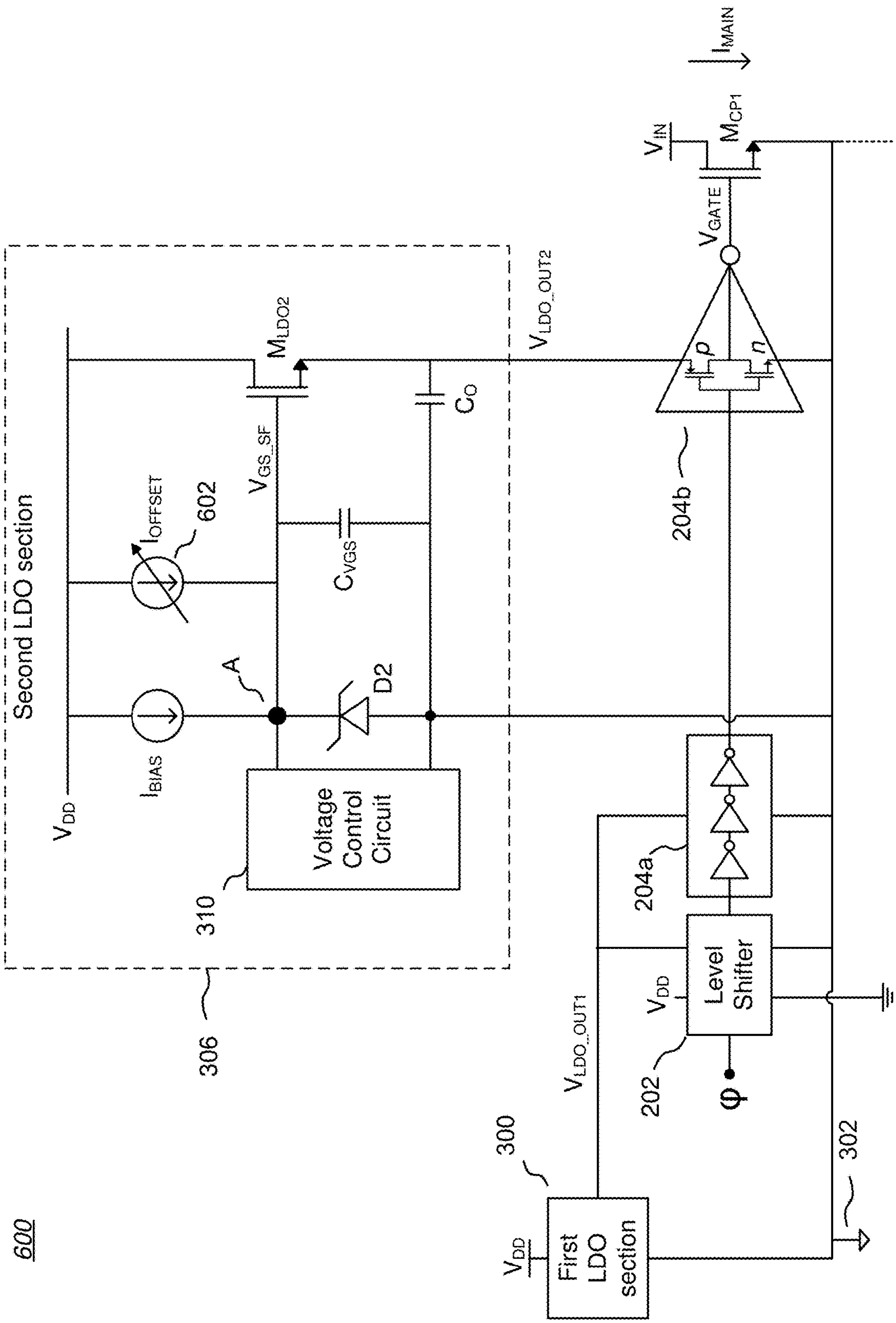

600
Second LDO section
306
310
Voltage Control Circuit
VDD
IBIAS
A
D2
IOFFSET
602
VGS_SF
CVGS
MLDO2
CO
VLDO_OUT2
VLDO_OUT1
First LDO section
300
VDD
Level Shifter
VDD
202
φ
204a
204b
p
n
VGATE
VIN
MCP1
IMAIN
302

FIG. 6A

620

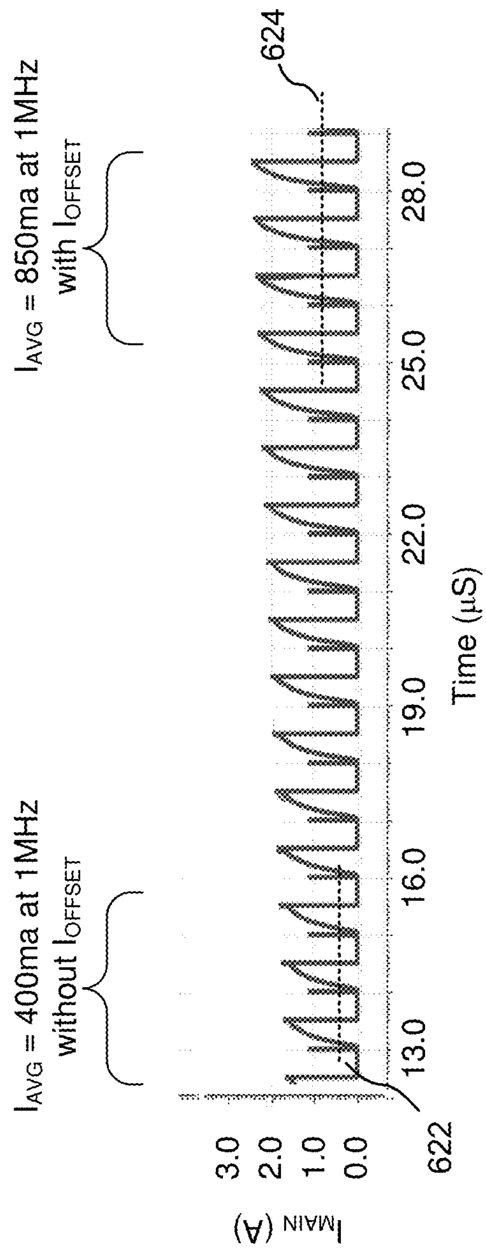

I_{AVG} = 400ma at 1MHz without I_{OFFSET}
I_{AVG} = 850ma at 1MHz with I_{OFFSET}
624
622
I_{MAIN} (A)
3.0
2.0
1.0
0.0
13.0
16.0
19.0
22.0
25.0
28.0
Time (μS)

FIG. 6B

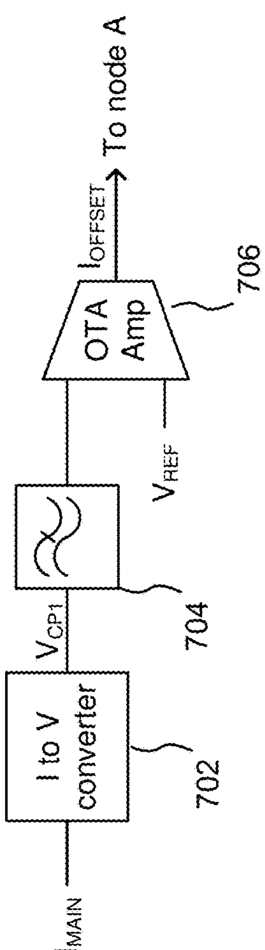


FIG. 7A

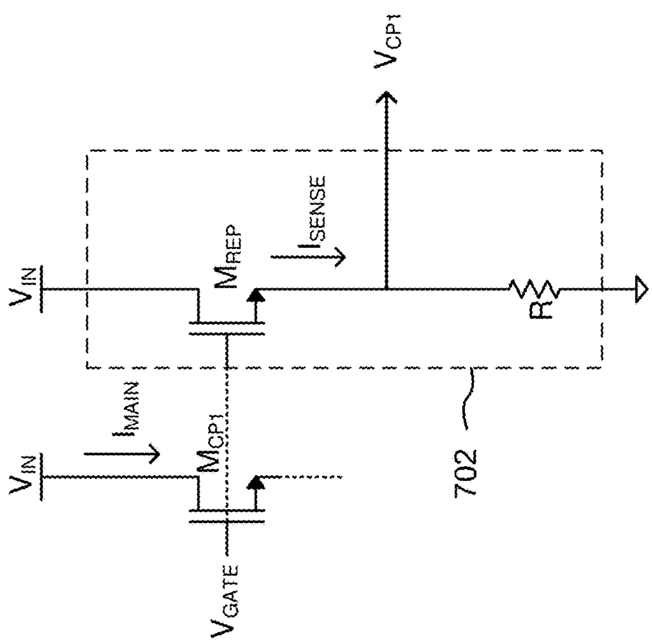


FIG. 7B

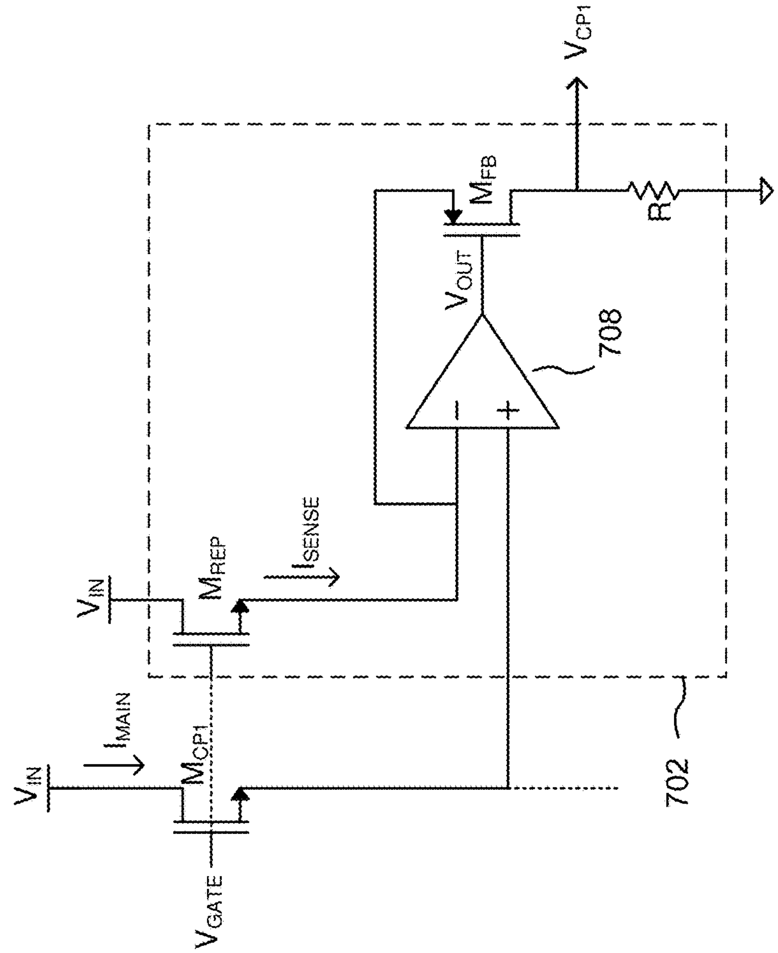


FIG. 7C

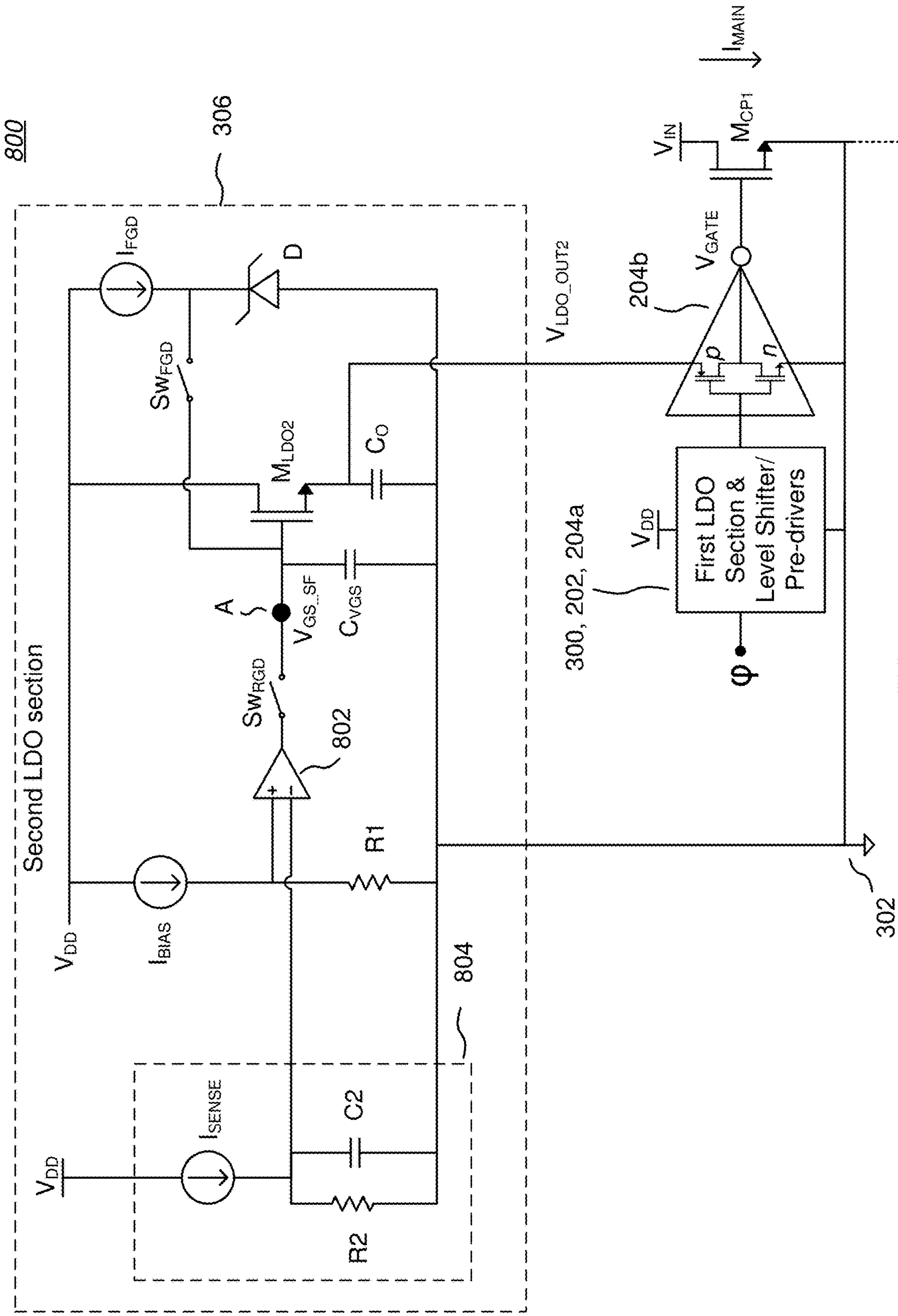

800
306
Second LDO section
V_{DD}
I_{BIAS}
I_{FGD}
D
Sw_{FGD}
M_{LDO2}
C_O
A
V_{GS_SF}
C_{VGS}
Sw_{RGD}
802
R1
804
V_{DD}
I_{SENSE}
C2
R2
V_{LDO_OUT2}
204b
p
n
V_{GATE}
V_{IN}
M_{CP1}
I_{MAIN}
300, 202, 204a
V_{DD}
First LDO Section & Level Shifter/ Pre-drivers
Φ
302

FIG. 8

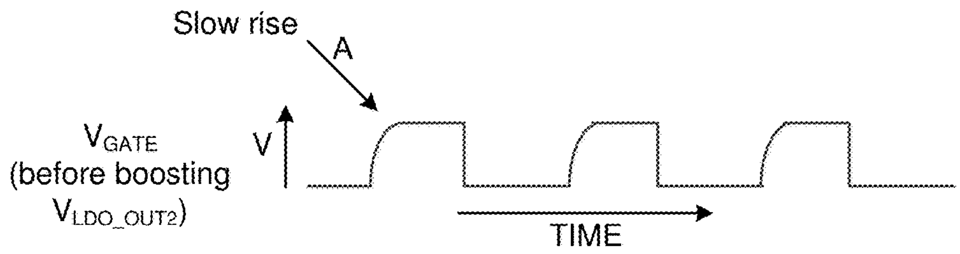


FIG. 9A

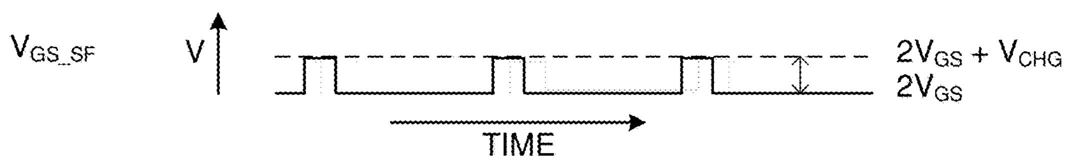


FIG. 9B

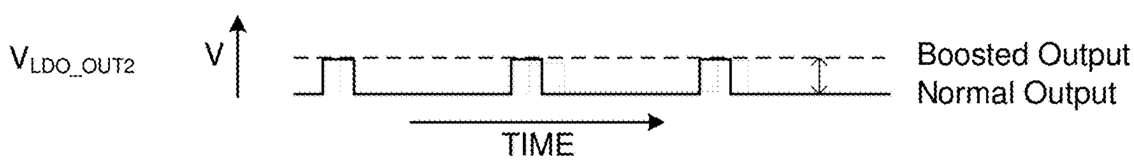


FIG. 9C

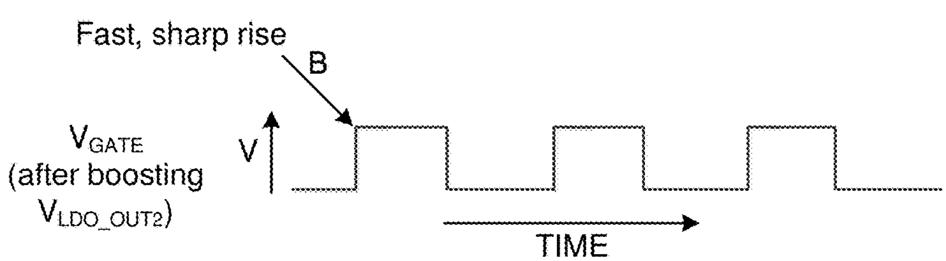


FIG. 9D

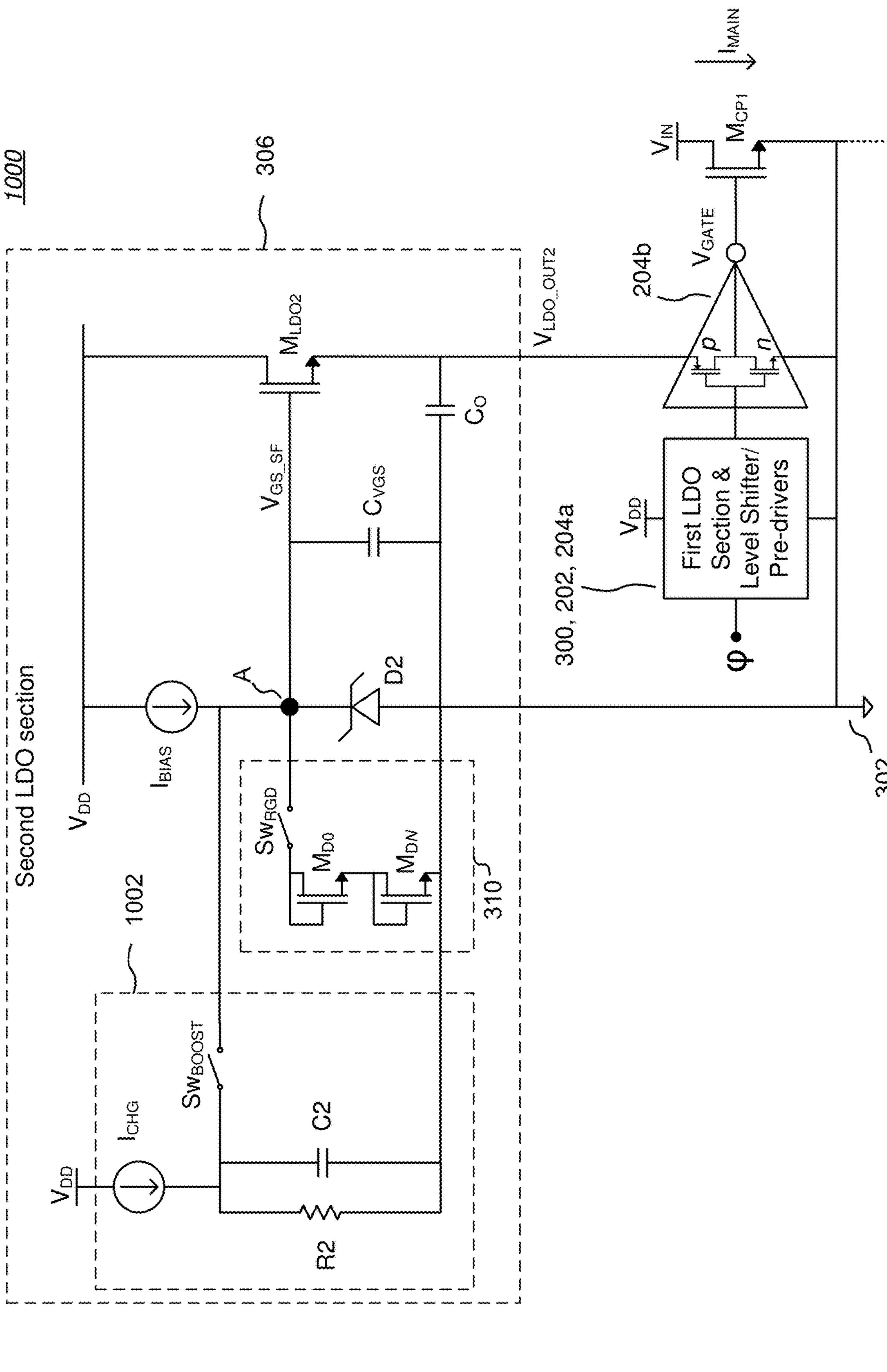

1000
306
Second LDO section
V_{DD}
I_{BIAS}
A
D2
M_{LDO2}
V_{GS_SF}
C_{VGS}
C_O
V_{LDO_OUT2}
Sw_{RGD}
M_{D0}
M_{DN}
310
1002
Sw_{BOOST}
I_{CHG}
V_{DD}
C2
R2
300, 202, 204a
V_{DD}
First LDO Section & Level Shifter/ Pre-drivers
Φ
204b
p
n
V_{GATE}
V_{IN}
M_{CP1}
I_{MAIN}
302

FIG. 10

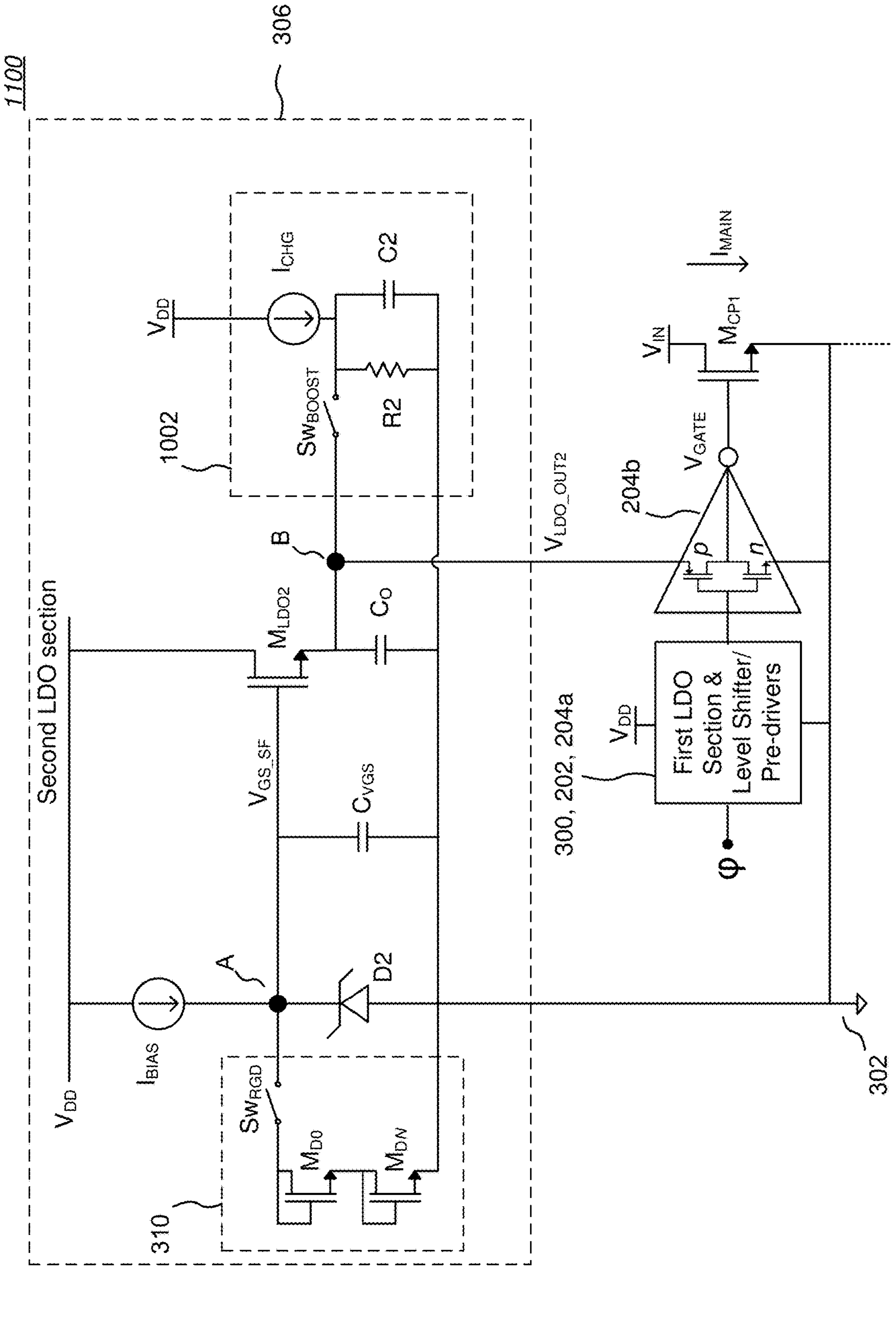

1100
306
Second LDO section
V_{DD}
I_{BIAS}
A
D2
Sw_{RGD}
M_{D0}
M_{DN}
310
V_{GS_SF}
C_{VGS}
M_{LDO2}
C_O
B
1002
V_{DD}
I_{CHG}
C2
Sw_{BOOST}
R2
V_{LDO_OUT2}
204b
V_{GATE}
V_{IN}
M_{CP1}
I_{MAIN}
p
n
300, 202, 204a
V_{DD}
First LDO Section & Level Shifter/ Pre-drivers
Φ
302

FIG. 11

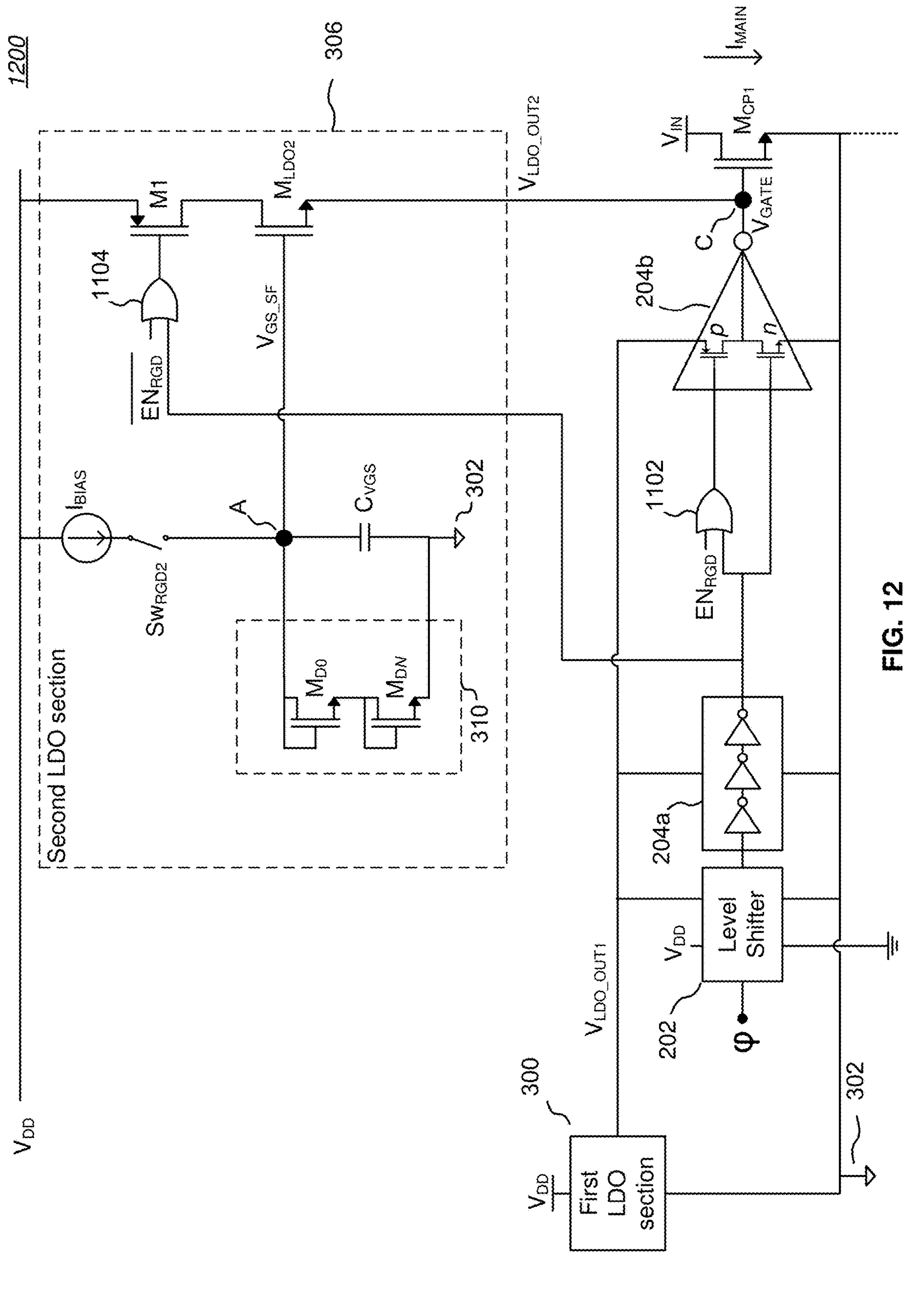

1200
306
Second LDO section
V_{DD}
I_{BIAS}
SW_{RGD2}
A
C_{VGS}
302
310
M_{D0}
M_{DN}
1104
$\overline{EN_{RGD}}$
M1
M_{LDO2}
V_{GS_SF}
V_{LDO_OUT2}
V_{IN}
M_{CP1}
I_{MAIN}
C
V_{GATE}
204b
p
n
1102
EN_{RGD}
204a
Level Shifter
V_{DD}
202
φ
V_{LDO_OUT1}
300
First LDO section
V_{DD}
302

FIG. 12

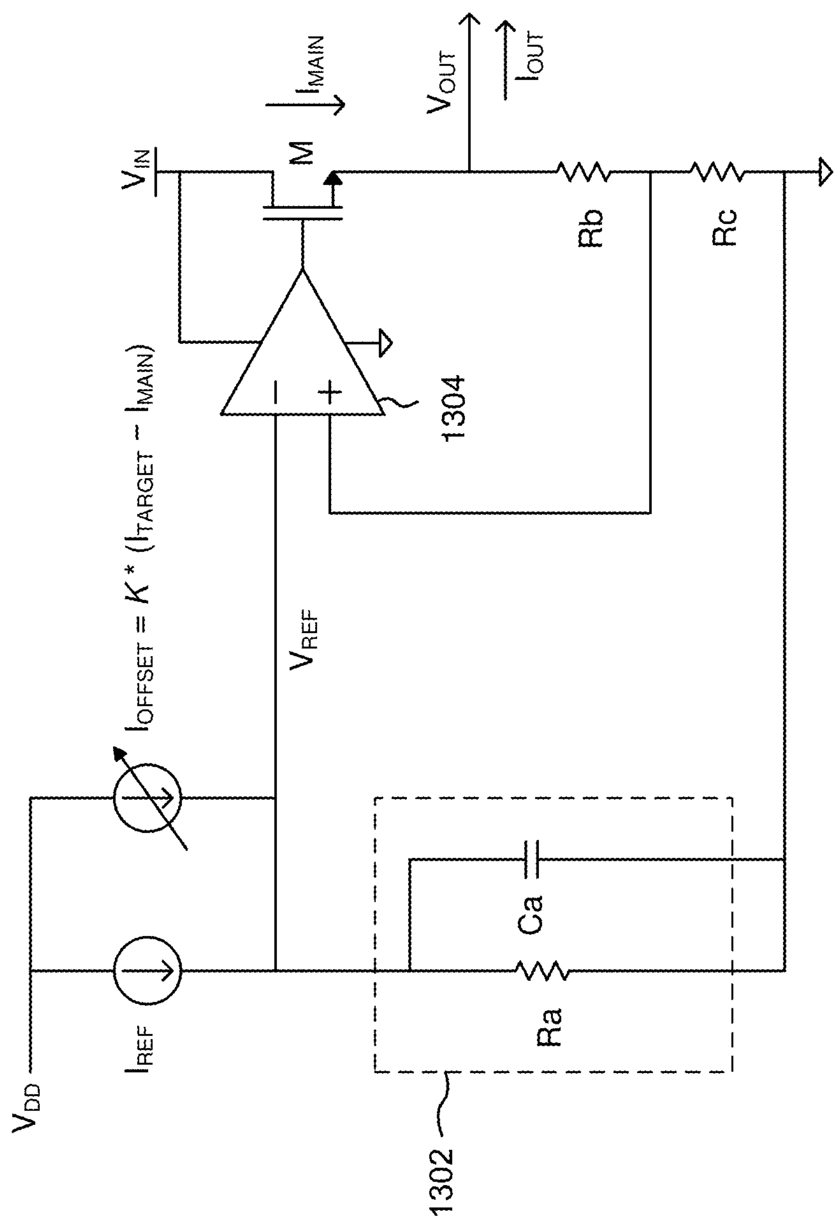

V_{DD}
I_{REF}
$I_{OFFSET} = K * (I_{TARGET} - I_{MAIN})$
V_{REF}
V_{IN}
M
I_{MAIN}
V_{OUT}
I_{OUT}
Rb
Rc
1304
Ca
Ra
1302

FIG. 13

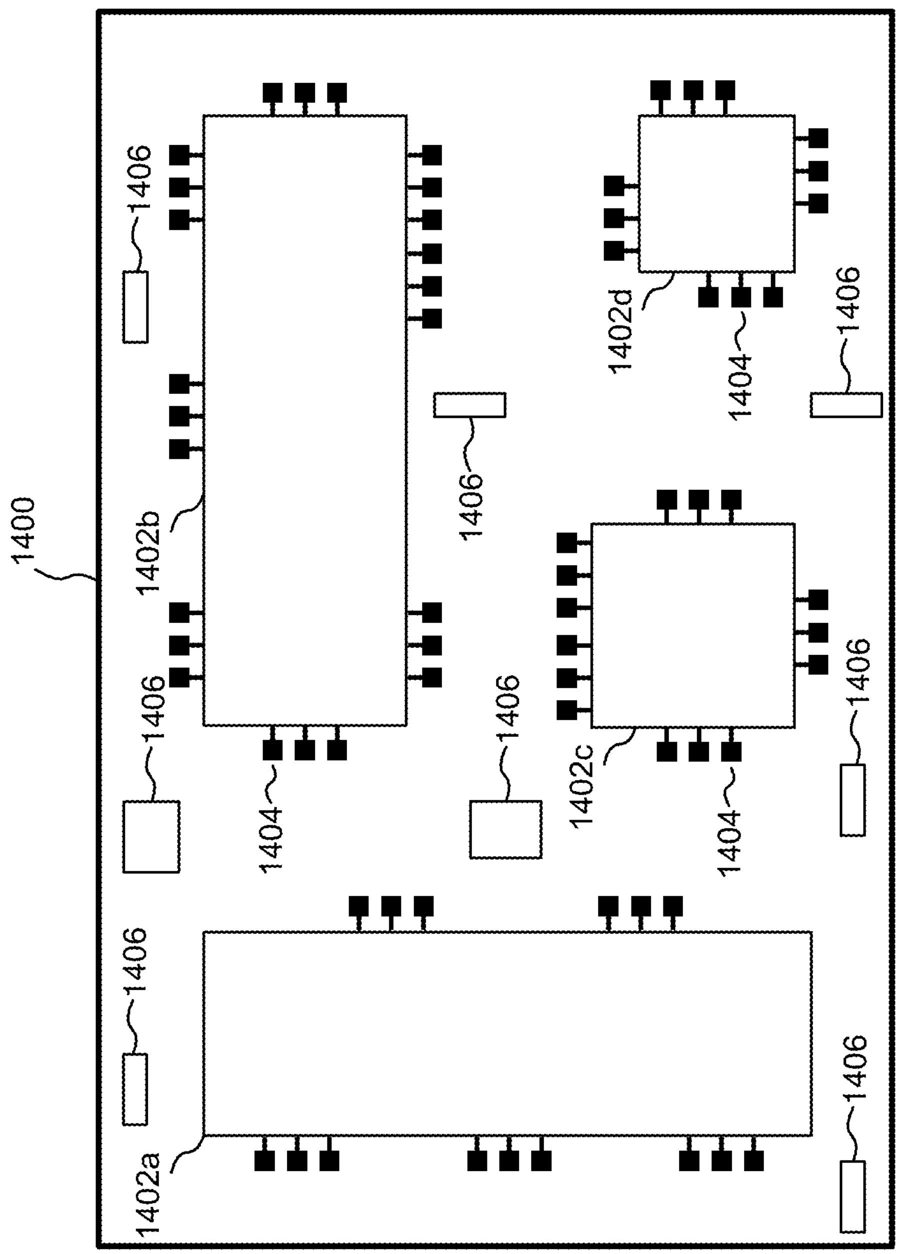

1400
1402a
1402b
1402c
1402d
1404
1404
1404
1406
1406
1406
1406
1406
1406
1406
1406

FIG. 14

1500

Controlling the ON resistance, R_{ON}, of a power FET in the power converter to lower the R_{ON} of the power FET in a first ON state during normal power converter operation and to raise the R_{ON} of the power FET in a second ON state to restrict flow of current through the power FET — 1502

Generating a feedback current proportional to a current flow through the power FET — 1504

Regulating the R_{ON} of the power FET in the second ON state in proportion to the generated feedback current to provide a reduced gate-drive to the power FET that is substantially independent of frequency — 1506

FIG. 15

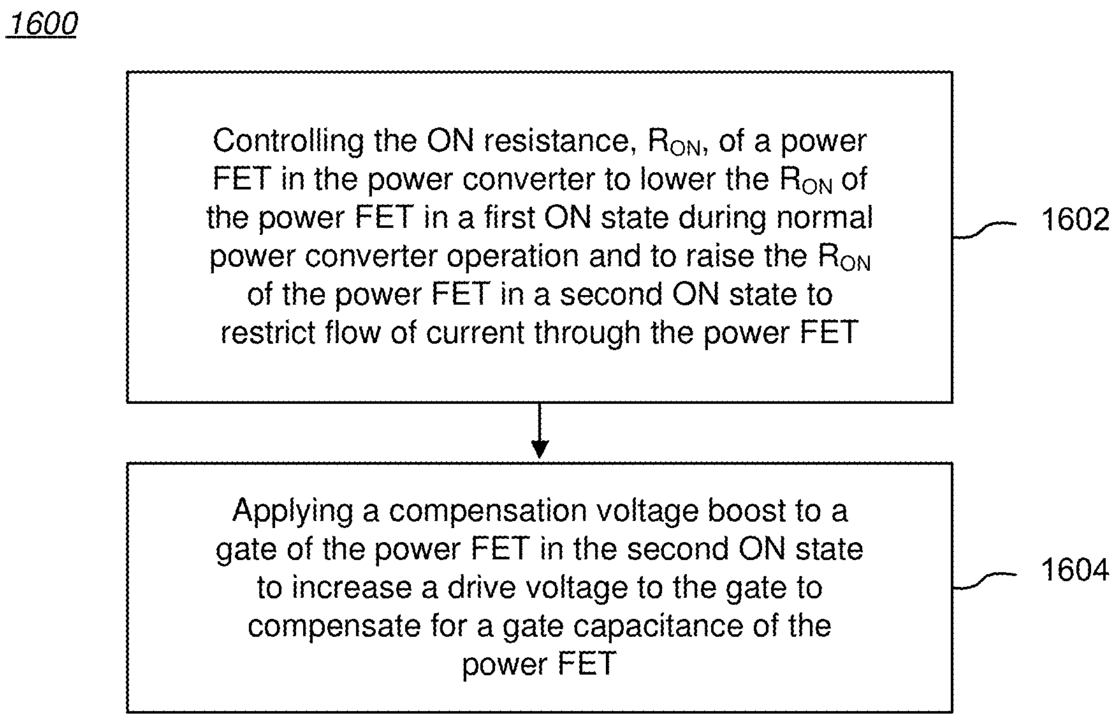

1600
Controlling the ON resistance, R_{ON}, of a power FET in the power converter to lower the R_{ON} of the power FET in a first ON state during normal power converter operation and to raise the R_{ON} of the power FET in a second ON state to restrict flow of current through the power FET
1602
Applying a compensation voltage boost to a gate of the power FET in the second ON state to increase a drive voltage to the gate to compensate for a gate capacitance of the power FET
1604

FIG. 16

1700

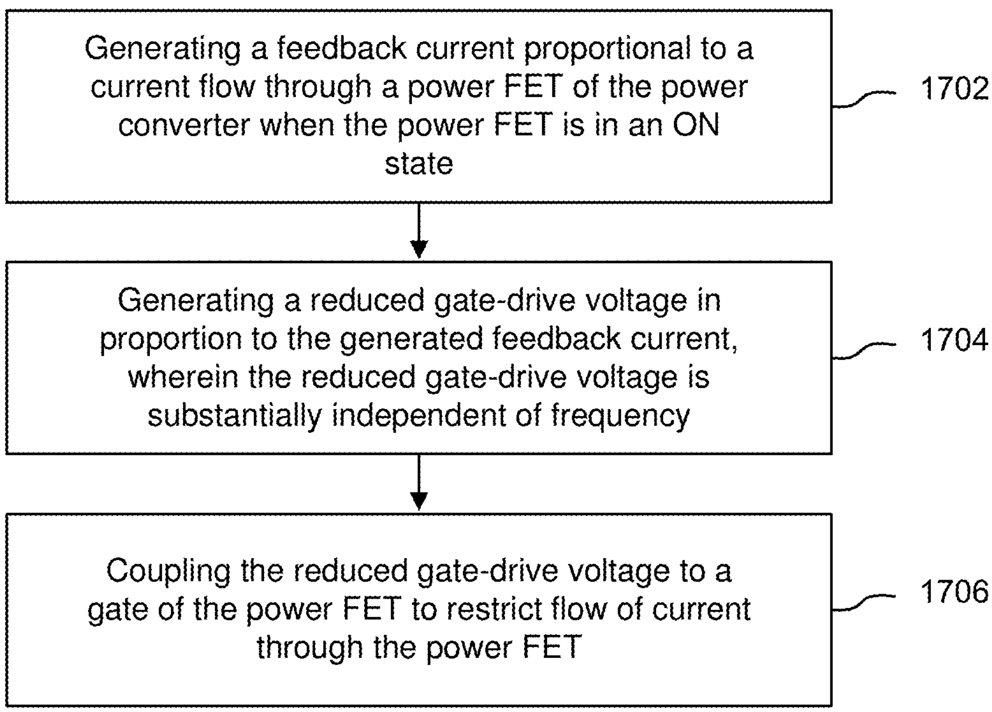

Generating a feedback current proportional to a current flow through a power FET of the power converter when the power FET is in an ON state
1702
Generating a reduced gate-drive voltage in proportion to the generated feedback current, wherein the reduced gate-drive voltage is substantially independent of frequency
1704
Coupling the reduced gate-drive voltage to a gate of the power FET to restrict flow of current through the power FET
1706

FIG. 17

ACCURATE REDUCED GATE-DRIVE CURRENT LIMITER

CROSS-REFERENCE TO RELATED APPLICATIONS-CLAIM OF PRIORITY

[0001] The present application is a continuation application of U.S. application Ser. No. 18/435,509, filed Feb. 7, 2024, for a "Accurate Reduced Gate-Drive Current Limiter", which is a continuation application of U.S. application Ser. No. 17/959,904, filed Oct. 4, 2022, for a "Accurate Reduced Gate-Drive Current Limiter", issued Mar. 19, 2024 as U.S. Pat. No. 11,936,371, which is herein incorporated by reference in its entirety. This invention may also be related to U.S. patent application Ser. No. 17/331,594, filed May 26, 2021, entitled "Dynamic Division Ratio Charge Pump Switching", the contents of which are herein incorporated by reference in its entirety.

BACKGROUND

(1) Technical Field

[0002] This invention relates to electronic circuits, and more particularly to power converter circuits, including DC-DC converter circuits, and current limiting circuits for such converter circuits.

(2) Background

[0003] Many electronic products, particularly mobile computing and/or communication products and components (e.g., notebook computers, ultra-book computers, tablet devices, LCD and LED displays) require multiple voltage levels. For example, radio frequency (RF) transmitter power amplifiers may require relatively high voltages (e.g., 12V or more), whereas logic circuitry may require a low voltage level (e.g., 1-2V). Still other circuitry may require an intermediate voltage level (e.g., 5-10V).

[0004] Power converters are often used to generate a lower or higher voltage from a common power source, such as a battery. One type of power converter comprises a converter circuit (e.g., a charge pump based on a switch-capacitor network), control circuitry, and, in some embodiments, auxiliary circuitry such as bias voltage generator(s), a clock generator, a voltage regulator, a voltage control circuit, etc. As used in this disclosure, the term "charge pump" refers to a switched-capacitor network configured to boost or buck V_{IN} to V_{OUT}. Examples of such charge pumps include cascade multiplier, Dickson, Ladder, Series-Parallel, Fibonacci, and Doubler switched-capacitor networks, all of which may be configured as a multi-phase or a single-phase network. Switched-capacitor network DC-DC converters are generally integrated circuits (ICs) that may have some external components (such as capacitors) and in most cases are characterized as having a fixed V_{IN} to V_{OUT} conversion ratio (e.g., division by 2 or by 3). An AC-DC power converter can be built up from a DC-DC power converter by, for example, first rectifying an AC input to a DC voltage and then applying the DC voltage to a DC-DC power converter.

[0005] To provide greater flexibility to system designers, and to deal with applications where a power source may change that requires different conversion ratios (e.g., as a battery discharges and outputs a lower voltage, or when the power source to a device is switched between a battery and an AC-DC power line source), it is useful to utilize a DC-DC power converter having a selectable conversion ratio. For example, U.S. Pat. No. 10,263,514 B1, issued Apr. 16, 2019, entitled "Selectable Conversion Ratio DC-DC Converter", assigned to the assignee of the present invention and hereby incorporated by this reference, describes a Dickson DC-DC power converter that may be switched between a divide-by-2 mode of operation and a divide-by-3 mode of operation. As another example, U.S. Pat. No. 9,203,299 B2, issued Dec. 1, 2015, entitled "Controller-Driven Reconfiguration of Switched-Capacitor Power Converter", now assigned to the assignee of the present invention and hereby incorporated by this reference, describes other DC-DC power converter architectures having reconfigurable conversion ratios.

[0006] A general problem with many FET-based DC-DC power converter architectures is that excessive current in-rush needs to be avoided during startup of the power converter. For example, for a selectable conversion ratio DC-DC converter of the type shown in U.S. Pat. No. 10,263,514 B1, absent sufficient guard circuitry, when an input voltage V_{IN} is first applied, none of the capacitors (sometimes known as "fly capacitors") would be charged initially and accordingly current rushes into the circuit. For example, if the ON resistance, R_{ON}, of the FET power switches is 1 milliohm (0.001 ohms), and V_{IN} is 10V, then as a result of Ohm's law, V=I×R, the in-rush current spikes at about 10,000 amps. In integrated circuit implementations, parasitic inductances exist (for example, due to on-die conductor routing and printed circuit board conductor routing) which transform a current spike to a voltage spike in accordance with inductor theory: V=L×dl/dt. Such voltage spikes electrically overstress the charge pump power switches, affecting their reliability, potentially to destruction. For a 1 ns 100 A pulse to generate 10V across the charge pump power switches, the parasitic inductance need only be about 100 pH. The resulting 10V spike may exceed the breakdown voltage of many of the FET switches, and of course, a larger current spike results in a larger voltage spike for the same parasitic inductance.

[0007] A related problem occurs when the fly capacitors of a DC-DC power converter are out of balance, meaning that a charge difference exists between fly capacitors connected by power switches. If charge balance is not maintained, current spikes and resulting damaging voltage spikes can occur.

[0008] Accordingly, it would be useful to be able to mitigate or eliminate what may be characterized as "potentially damaging events" in power converters (e.g., damaging current spikes that may occur for a variety of reasons, including in-rush current, charge transfer current, short circuits, EMI events, and the like).

SUMMARY

[0009] The present invention provides circuits and methods that limit current through power FETs of power converters, thereby mitigating or eliminating potentially damaging events, independent of converter switching frequency, device mismatches, and process, voltage, and temperature (PVT) variations. Such circuits and methods provide protection against potentially damaging events such as current spikes during a "soft-start" for power converters and during dynamic charge balancing, without requiring added circuitry directed to those functions.

[0010] Embodiments of the present invention may some or all of the following advantages: a current limiting value that is independent of a power converter switching frequency, device mismatches and process, voltage, and/or temperature (PVT) variations; accurate soft-start current limiting; reliable operation of power FETs in power converters having reconfigurable conversion ratios when dynamically changing conversion ratios; and the ability to keep a level output voltage when a power converter operates at full load in different operating and PVT conditions.

[0011] In contrast to open-loop implementations of an LDO power supply, embodiments of the present invention utilize an added closed-loop feedback circuit and/or an added calibrated compensation circuit to closely control the control voltage V_{GATE} applied to the gate of a power FET in a manner that is substantially independent of frequency. Depending on the nature of the added circuit, connection to the LDO power supply may be made at one of several nodes. Connecting a feedback or compensation circuit to the gate of an LDO source-follower FET allows the gate voltage of the source-follower FET to be regulated to control the LDO output voltage to a final inverter coupled to the gate of a power FET so that V_{GATE} is adjusted to provide a reduced gate-drive to the power FET. Connecting a feedback or compensation circuit at the output of the LDO power supply allows the LDO output voltage to the final inverter to be directly regulated so that V_{GATE} is adjusted to provide a reduced gate-drive to the power FET. Connecting a feedback or compensation circuit to the gate of the power FET allows V_{GATE} to be directly set to provide a reduced gate-drive to the power FET.

[0012] The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF THE DRAWINGS

[0013] FIG. **1** is a block diagram of one embodiment of a DC-DC selectable conversion ratio power converter.

[0014] FIG. **2** is a block diagram of one section of a power converter showing implementation details for the primary switch of one section.

[0015] FIG. **3** is a schematic diagram of one embodiment of the LDO power supply of FIG. **2**.

[0016] FIG. **4**A is a graph showing the gate voltage V_{GS} for the FET M_{CP1} and the current I_{MAIN} through FET M_{CP1} as a function of time at a switching frequency of kHz in a reduced gate-drive mode of operation.

[0017] FIG. **4**B is a graph showing the gate voltage V_{GS} for FET M_{CP1} and the current I_{MAIN} through FET M_{CP1} as a function of time at a switching frequency of 1 MHz in a reduced gate-drive mode of operation.

[0018] FIG. **5** is a block diagram of an LDO power supply circuit showing possible connections nodes A, B, and C for added feedback or compensation circuits.

[0019] FIG. **6**A is a block diagram of a first embodiment of an enhanced LDO power supply utilizing a closed-loop direct supplementation of V_{GS_SF}.

[0020] FIG. **6**B is a graph showing the current I_{MAIN} through FET M_{CP1} as a function of time at a switching frequency of 1 MHz in a reduced gate-drive mode of operation.

[0021] FIG. **7**A is a block diagram of an example circuit for generating the supplemental feedback offset current I_{OFFSET}.

[0022] FIG. **7**B is a schematic diagram of a first example embodiment of a current-to-voltage converter.

[0023] FIG. **7**C is a schematic diagram of a second example embodiment of a current-to-voltage converter.

[0024] FIG. **8** is a block diagram of a second embodiment of an enhanced LDO power supply.

[0025] FIG. **9**A is a graph of V_{GATE} as a function of time without the benefit of a compensation voltage boost.

[0026] FIG. **9**B is a graph of V_{GS_SF} as a function of time, showing the periodic augmentation to the normal $2V_{GS}$ voltage that a compensation voltage boost V_{CHG} provides if applied at Node A.

[0027] FIG. **9**C is a graph of V_{LDO_OUT2} as a function of time, showing the periodic augmentation in the normal reduced gate-drive output voltage resulting from application of an appropriately scaled compensation voltage boost V_{CHG} to Node A or to Node B (see FIG. **5**).

[0028] FIG. **9**D is a graph of V_{GATE} as a function of time with the benefit of a compensation voltage boost to V_{LDO_OUT2}.

[0029] FIG. **10** is a block diagram of a third embodiment of an enhanced LDO power supply.

[0030] FIG. **11** is a block diagram of a fourth embodiment of an enhanced LDO power supply.

[0031] FIG. **12** is a block diagram of a fifth embodiment of an enhanced LDO power supply.

[0032] FIG. **13** is a schematic diagram of an LDO.

[0033] FIG. **14** is a top plan view of a substrate that may be, for example, a printed circuit board or chip module substrate (e.g., a thin-film tile).

[0034] FIG. **15** is a process flow chart showing a first method for protecting a power converter.

[0035] FIG. **16** is a process flow chart showing a second method for protecting a power converter.

[0036] FIG. **17** is a process flow chart showing a third method for protecting a power converter.

[0037] Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0038] The present invention provides circuits and methods that limit current through power FETs of power converters, thereby mitigating or eliminating potentially damaging events, independent of converter switching frequency, device mismatches, and process, voltage, and temperature (PVT) variations. Such circuits and methods provide protection against potentially damaging events such as current spikes during a "soft-start" for power converters and during dynamic charge balancing, without requiring added circuitry directed to those functions.

Example Selectable Conversion Ratio Power Converter

[0039] FIG. **1** is a block diagram of one embodiment of a DC-DC selectable conversion ratio power converter **100**. The specific illustrated power converter **100** may be selectably configured to be either a divide-by-2 Dickson converter or a divide-by-3 Dickson converter using the same basic circuit. The same power converter **100** may be used for DC-to-DC boost conversion by reversing the voltage input and voltage output. The illustrated power converter **100** includes two parallel sections **102**_a_, **102**_b_ that are coupled between a voltage source V_{IN} and a reference potential **104** such as circuit ground. Each section **102**_a_, **102**_b_ includes 3 series-connected switches **S1**-**S3** coupled in series to a first branch comprising 2 series-connected switches **S4**-**S5**, and to a second branch comprising 2 series-connected switches **S6**-**S7**. Each switch may comprise, for example, one or more FETs, including one or more MOSFETs.

[0040] In each section **102***a*, **102***b*, coupled between a first upper pair of switches **S1**, **S2** and a first branch pair of switches **S4**, **S5** is a first capacitor C**1***a*, C**1***b*, and coupled between a second upper pair of switches **S2**, **S3** and a second branch pair of switches **S6**, **S7** is a second capacitor C**2***a*, C**2***b*. Depending on the output ratio configuration (divide-by-2 or divide-by-3), each section **102***a*, **102***b* generates an output voltage at a node V_X that is coupled to an output capacitor C_{OUT}.

[0041] At least some of the switches **S1**-**S7** may be selectively controlled to be in an ON (conductive) or OFF (blocking) state by control circuitry (not shown). At least some of the switches **S1**-**S7** may be selectively coupled to one of two non-overlapping complementary clock phases, P**1** or P**2**. Some of the switches **S1**-**S7** may be permanently coupled to one of the two complementary clock phases, P**1** or P**2**. TABLE 1 below shows the configuration of the state or associated clock phase for each of the switches **S1**-**S7** of the two parallel sections **102***a*, **102***b* for both a divide-by-2 configuration and a divide-by-3 configuration.

TABLE 1

Divide-by-2 Configuration		Divide-by-3 Configuration	
Section 102a	Section 102b	Section 102a	Section 102b
S1 = P1	S1 = P2	S1 = P1	S1 = P2
S2 = ON	S2 = ON	S2 = P2	S2 = P1
S3 = P2	S3 = P1	S3 = P1	S3 = P2
S4 = P1	S4 = P2	S4 = P1	S4 = P2
S5 = P2	S5 = P1	S5 = P2	S5 = P1
S6 = P1	S6 = P2	S6 = P2	S6 = P1
S7 = P2	S7 = P1	S7 = P1	S7 = P2

[0042] Note that the clock phase associations for section **102***b* are the complements of the clock phase associations for section **102***a*. The complementary phasing of the two parallel sections **102***a*, **102***b* provides output ripple smoothing and additional current capacity. Additional sections may be included to provide even more current capacity. Complimentary pairs of additional sections may be controlled by clock signal phases that are 180° apart and that have a different phase than P1 or P2 to provide even more output ripple smoothing (e.g., 45° or 60°—or multiples of those values—out of phase with respect to P**1** and P**2**).

[0043] In a FET-based implementation, ON/OFF control signals or clock phase signals are coupled to the gate of each switch **S1**-**S7** through at least a driver circuit, and in many cases through both a level shifter circuit and a driver circuit (see FIG. **2** for more details).

[0044] In either configuration, the non-overlapping complementary clock signals P**1**, P**2** open or close associated power switches, causing charge to be transferred from the fly capacitors C**1***a*, C**1***b*, C**2***a*, C**2***b* into C_{OUT}, resulting in a voltage across C_{OUT} of V_{IN}/X, where X=2 or 3. Further details of the operation of this and similar DC-DC selectable conversion ratio power converters are set forth in U.S. Pat. No. 10,263,514 B1.

Limiting In-Rush Current During Soft-Starts

[0045] As noted above, damaging current spikes in power converters may occur for a variety of reasons, including in-rush current, charge transfer current, short circuits, and the like. For example, with respect to DC-DC power converters having selectable conversion ratios, switching from one conversion ratio (e.g., divide-by-2, or "DIV2") to another conversion ratio (e.g., divide-by-3, or "DIV3") may result in a charge imbalance across the fly capacitors, resulting in potentially damaging in-rush currents. Accordingly, a common practice for avoiding potentially damaging events has been to switch the DC-DC power converter OFF, allow the fly capacitors to discharge, change the conversion ratio configuration (e.g., by changing clock phasing to the switches **S1**-**S7** as needed), and turn the power back ON, relying on conventional startup circuitry to mitigate in-rush current spikes. A disadvantage of this practice is that the process can take several milliseconds to complete and cannot be completed under load.

[0046] One aspect of the present invention encompasses circuits and method for mitigating or eliminating potentially damaging events if they occur or are to occur (e.g., are known in advance, as when a conversion ratio is to be dynamically changed). Mitigating or eliminating potentially damaging events enables switching selectable conversion ratios DC-DC power converters from one conversion ratio to another conversion ratio under load without turning off the power converter circuitry or suspending switching of the charge pump power switches.

[0047] As described in U.S. patent application Ser. No. 17/331,594, it is desirable, and often necessary, to limit the current drawn by a power converter from an input supply during startup to avoid high in-rush currents, particularly when dynamically changing the conversion ratio of the power converter. Referring to FIG. **1**, a conceptual solution during a "soft-start" period before steady-state operation of the power converter **100** is to enable current sources **106***a*, **106***b*, respectively coupled to the sections **102***a*, **102***b*, and operate the switches **S1**-**S7** so as to allow the fly capacitors C**1***a*, C**1***b*, C**2***a*, C**2***b* to be charged by the current sources **106***a*, **106***b* to a sufficient state to prevent current spikes. More specifically, the current sources **106***a*, **106***b* may be coupled between V_{IN} and a node between switches **S1** and **S2** that also couples to a respective fly capacitor C**1***a*, C**1***b*. Closing switch **S2** during the soft-start period also couples the current sources **106***a*, **106***b* to a respective fly capacitor C**2***a*, C**2***b*.

[0048] One way to implement the current sources **106***a*, **106***b* is by means of a power switch coupled in series with a resistor between V_{IN} and the node between switches **S1** and **S2** (the power switch and resistor are thus coupled in parallel with switch **S1**). Closing the power switch creates a high resistive path during the soft-start period to limit the start-up current. Once the power converter output voltage V_{OUT} nears its regulated value, the power switch is opened. However, such an implementation requires a very large FET for the power switch in order to withstand the voltage and current at startup, and thus requires a large amount of integrated circuit (IC) die area (as much as about 25% in some embodiments).

[0049] In U.S. patent application Ser. No. 17/331,594, it was realized that switch **S1** and its driver circuitry could be adapted to perform the functions of the current sources **106***a*, **106***b*, thus eliminating the need for additional large power switches and resistors. In particular, it was realized that the power converter switches S1-S7 are normally operated in an "over-driven" or "full drive" condition when set to an ON (conducting) state. An overdriven FET gate creates a stronger conduction channel, effectively lowering the ON resistance, R_{ON}, of the FET. With that insight, it was further realized that increasing R_{ON} for some or all of the power FETs in a power converter (especially switch S1) during potentially damaging events (e.g., during startup or when dynamically re-configuring the conversion ratio of the power converter) would reduce current flow through the FETs and thus protect against excessive current spikes.

[0050] FIG. **2** is a block diagram of one section **102***a* of a power converter **200** showing implementation details for the primary switch of one section. While section **102***a* is shown, a similar configuration exists for section **102***b*. In the illustrated example, switch S1 is implemented as a FET M_{CP1}. A clock phase (P1 in this example) is coupled through a level shifter **202** to the input of a driver circuit **204**. The output of the driver circuit **204** is coupled to the gate of M_{CP1}. In the illustrated example, a low-dropout (LDO) power supply **206** provides power to the level shifter **202** and the driver circuit **204**. As described in U.S. patent application Ser. No. 17/331,594 referenced above, the LDO power supply **206** can selectively increase R_{ON} for a power FET in a power converter by actively controlling the driver voltage to the gate of the power FET. During normal power converter operation, the power FET driver voltage may be set to overdrive the FET gate to lower R_{ON} to a desired level that allows high current flow for a particular application. However, for other scenarios (e.g., during soft-startup), the power FET driver voltage may be reduced so as to increase R_{ON} and thus impede current flow through the power FET to a desired level.

LDO Power Supply Embodiment with Reduced Gate-Drive Capability

[0051] FIG. **3** is a schematic diagram of one embodiment of the LDO power supply **206** of FIG. **2**. As in FIG. **2**, the LDO power supply **206** provides power to a level shifter **202** and to a driver **204** coupled to the gate of an associated power FET M_{CP1}. The input to the gate control circuit, q (e.g., either clock signal P**1** or clock signal P**2**, or an ON or OFF control signal), is applied to the input of the level shifter **202**. The level shifter **202** translates the input signal from one voltage domain (e.g., digital logic voltages) to another voltage domain (e.g., transistor control voltages). The output of the level shifter **202** thus follows the input signal but in a different voltage range.

[0052] The output of the level shifter **202** is coupled to the input of a driver circuit **204**, the output of which is coupled to the gate of FET M_{CP1}. In the illustrated example, the driver circuit **204** includes a pre-driver **204***a* (comprising a set of three series-coupled inverters in this example) and a series-coupled final inverter **204***b*. Internally, the final inverter **204***b* has at least one NMOS FET n and one PMOS FET p with coupled conduction channels, drain-to-drain, with each FET n, p having a gate driven by the output of the pre-driver **204***a*. The drains of PMOS FET p and of NMOS FET n are coupled to the gate of power FET M_{CP1}.

[0053] In some embodiments, the inverters may increase in physical size from inverter to inverter in order to provide sufficient current drive capability to charge the gate of FET M_{CP1}. For example, in a driver circuit **204** having three series-coupled inverters in the pre-driver **204***a*, the first inverter may have a relative size of "1", the second inverter may be 3 times larger than the first inverter, and the third inverter may be 9 times larger than the first inverter. Lastly, the final inverter **204***b* may be 27 times larger than the first inverter in the pre-driver **204***a*. The multipliers for the stages may differ from the 1×, 3×, 9×, and 27× ratios, although generally each stage is larger than the previous one to avoid having very slow rising and falling edges. In alternative embodiments, the number of inverter stages may be fewer or greater, and non-inverting stages (buffer amplifiers) may be used rather than inverting stages. Accordingly, the illustrated driver circuit **204** is exemplary only, and other circuits may be used to couple the output of the level shifter **202** to the gate of FET M_{CP1}.

[0054] Power to the level shifter **202** and the driver circuit **204** is provided by LDO power supply **206**. In the illustrated example, the power source for the level shifter **202** and the pre-driver **204***a* is provided by a first LDO section **300**. The first LDO section **300** comprises a source follower (common drain) amplifier circuit that includes a regulated FET M_{LDO1} having its conduction channel (between drain and source) coupled in series with a resistor R**1** between a supply voltage, V_{DD}, and a floating reference potential **302**. As an example, the supply voltage V_{DD} may be V_{IN} to one phase of the charge pump that includes FET M_{CP1} or may be coupled to the voltage output from another phase of the charge pump-basically, any voltage than is sufficiently high and has sufficient drive strength for the circuit. A current source **304** is coupled in series with a Zener diode D**1** between V_{DD} and the reference potential **302**. A current source may be configured from transistors and/or diodes using a variety of circuits. The output of the current source **304** before the Zener diode D**1** provides an essentially constant bias voltage to the gate of FET M_{LDO1}. The bias current flows through the Zener diode D**1** and ensures that the diode is always in reverse bias. The source of the FET M_{LDO1} provides a drive voltage V_{LDO_OUT1} to the level shifter **202** and the pre-driver **204***a*.

[0055] Unlike a conventional diode that blocks any flow of current through itself when reverse biased, as soon as the reverse voltage reaches a pre-determined value, a Zener diode begins to conduct. This applied reverse voltage remains almost constant even with large changes in current (so long as the current remains between a breakdown minimum current and a maximum current rating for the Zener diode). A Zener diode continues to regulate its voltage until the holding current of the diode falls below the minimum current value in the reverse breakdown region.

[0056] The final inverter **204***b* is powered by a second LDO section **306** that includes a source-follower FET M_{LDO2} having its conduction channel (between drain and source) coupled between the supply voltage V_{DD} and the final inverter **204***b*. The gate of FET M_{LDO2} is coupled to a separate gate driver circuit that is independent of the gate driving circuitry for FET M_{LDO1}. The principal function of the gate driver circuit of the second LDO section **306** is to enable at least two different voltage levels at Node A to be coupled to the gate of FET M_{LDO2}, which in turn determines the output voltage level V_{GATE} provided by the final inverter **204***b* driving the associated power FET M_{CP1}. The associated power FET M_{CP1} thus can be placed into (1) an overdriven or "full gate-drive" ON state having low R_{ON} for normal power converter operation, or (2) at least one current-limiting reduced gate-drive ON state having a higher

R_{ON} selected to provide protection against potentially damaging events (e.g., in-rush or charge transfer current), such as during dynamic re-configuration of the conversion ratio of the power converter, during power converter startup, when balancing charge among fly capacitors within the power converter, or during fault events such as short circuit events.

[0057] The gate driver circuit for FET M_{LDO2} includes a current source **308** coupled in series with a Zener diode D**2** between V_{DD} and the reference potential **302**. The gate of FET M_{LDO2} is coupled to Node A between the current source **308** and the Zener diode D**2**. The output I_{BIAS} of the current source **308** before the Zener diode D**2** at Node A provides an essentially constant bias voltage Vas SE to the gate of FET M_{LDO2}. The source of the FET M_{LDO2} provides a drive voltage V_{LDO_OUT2} to the final driver **204***b*.

[0058] In parallel with the Zener diode D**2** is a voltage control circuit **310** comprising a reduced gated-drive switch Sw_{RGD} series-coupled to a first diode-connected FET M_{D0} and at least one additional diode-connected FET M_{DN}, where N>1. As illustrated, one terminal of the switch Sw_{RGD} is coupled to Node A, and one terminal of the additional diode-connected FET M_{DN} is coupled to the floating reference potential **302**. Note that the switch Sw_{RGD} may be positioned anywhere along the voltage control circuit **310** to interrupt or enable current flow through that circuit. For example, the order of switch Sw_{RGD} and FETs M_{D0} and M_{DN} from Node A the floating reference potential **302** may be (1) Sw_{RGD}, M_{D0}, M_{DN} (as illustrated), (2) M_{D0}, Sw_{RGD}, M_{DN}, or (3) M_{D0}, M_{DN}, Sw_{RGD}. However, positioning the switch Sw_{RGD} as shown in FIG. **3** may reduce parasitic influences on FET M_{LDO2} due, for example, to the capacitances of FETs M_{D0} and/or M_{DN}.

[0059] A decoupling capacitor C_O is coupled between the source of FET M_{LDO2} and the floating reference potential **302**. A capacitor C_{VGS} is coupled between the gate of FET M_{LDO2} and the floating reference potential **302** to maintain essentially a constant voltage across the Zener diode D2 when the floating reference potential **302** switches voltages.

[0060] A function of the diode-connected FET M_{D0} is to offset FET M_{LDO2}, since the threshold voltages of FET M_{D0} and FET M_{LDO2} effectively cancel. A function of the additional diode-connected FETs M_{DN} is to set the current I_{MAIN} through FET M_{CP1} in proportion to the ratio of the sizes of FET M_{CP1} to FET M_{DN} when switch Sw_{RGD} is CLOSED and the current mirror function of the voltage control circuit **310** is engaged. More particularly, the current I_{MAIN} through FET M_{CP1} is proportional to the current I_{BIAS} from the current source **308** and the size ratio of FET MIN to FET M_{CP1}. For example, if the current source **308** output is 1 mA, and FET M_{CP1} is 1,000 times the size of FET M_{DN} (W/L M_{CP1}=1000×W/L M_{DN}), then the maximum current through FET M_{CP1} is 1,000×1 mA=1 A. This is achieved by ensuring the gate-to-source voltage V_{GS} of FET M_{DN} is the same as that of FET M_{CP1}. The maximum gate voltage of FET M_{CP1} is the voltage at Node A minus the threshold voltage V_{TH} of FET M_{LDO2}. Including FET M_{D0} increases the voltage at Node A by a second threshold voltage V_{GS}, so the voltage at Node A=(Vas of FET M_{DN})+ (V_{TH} Of FET M_{D0}), or $2V_{GS}$. If FET M_{LDO2} and FET M_{D0} are matched (ratiometrically), then the maximum the V_{GS} of FET M_{CP1} can reach is the same as the V_{GS} of FET M_{DN}, and this equality tracks over process, temperature, etc.

[0061] As noted, the diode-connected FET(s) M_{DN} are ratioed in size with respect to FET M_{CP1}. In some embodiments, FETs M_{LDO1}, M_{LDO2}, M_{D0}, and M_{CP1} may be segmented FETs, meaning that a device intended to function as a large FET is fabricated as multiple (e.g., 10,000) small FETs coupled in parallel (the individual small FETs may be called "fingers", reflecting typical aspects of their physical layout on an IC die). The diode-connected FET(s) M_{D0}, M_{DN} may be fabricated using the same technology, but can be made with a much smaller number of FET fingers (e.g., as few as one finger). Because of the source-follower configuration of FET M_{LDO2} and the final inverter **204***b*, a small change in current flow through the voltage control circuit **310** affecting the voltage at the gate of FET M_{LDO2} causes a proportionally larger current flow I_{MAIN} through power FET M_{CP1} determined by the size ratio of FET M_{CP1} to FET M_{DN}.

[0062] Adding more than one diode-connected FET M_{DN} allows adjustment of the size ratio of FET M_{CP1} to FET M_{DN}. For instance, if FET M_{CP1} has a width of 100 and 1,000 fingers, then a first FET M_{DN} should also have a width of 100 to match (but may only have 1 finger). Hence the size ratio of FET M_{DN} to FET M_{CP1} is 1,000 to 1, and 1 mA from the current source **308** means 1 A through FET M_{CP1}. To change the size ratio to 2,000 to 1, two diode-connected FETs M_{DN} may be coupled in series (source to drain). If the FET width is still 100, the effective number of fingers of the two diode-connected FETs M_{DN} is one-half, giving a size ratio of 2,000 to 1 with respect to FET M_{CP1}.

[0063] As noted above, an important function of the gate driver circuit is that it provides a selectable amount of regulated gate bias voltage V_{GS_SF} to FET M_{LDO2}, which in turn controls the power supply to, and voltage output of, the final inverter **204***b*. When switch Sw_{RGD} is OPEN, then the voltage control circuit **310** is disconnected from Node A—and therefore from the gate of FET M_{LDO2}—and thus has essentially no effect on the output of FET M_{LDO2}; accordingly, the final inverter **204***b* can overdrive the gate of FET M_{CP1} to a selected level determined by the Zener diode D**2**.

[0064] When switch Sw_{RGD} is CLOSED-such as during startup of the power converter or when dynamically switching conversion ratios or rebalancing charge amount fly capacitors-then the voltage control circuit **310** operates as a bypass to divert current around diode D**2** and lower the voltage at Node A, thus reducing the drive voltage to FET M_{LDO2}. The reduced gate-drive voltage to FET M_{LDO2} in turn reduces the power to the final inverter **204***b* and thus reduces the gate-drive voltage V_{GATE} to the power FET M_{CP1}. If the drain-source voltage V_{DS} is high enough to cause the power FET M_{CP1} to be in saturation, then the power FET M_{CP1} acts as a controlled current-limited source. If V_{DS} is below the level that would cause the power FET M_{CP1} to be in saturation, then the power FET M_{CP1} should be in its linear range of operation with an increased R_{ON} value compared to the R_{ON} value when in a normal overdriven state. In either case-saturation-mode controlled current-limited source or linear-mode increased R_{ON}—at least some of the power FETs of a power converter may limit the current in the FETs and therefore inhibit excessive current spikes, thus protecting the power FETs (as well as other coupled circuitry) from large voltage spikes. Selectively varying the I_{BIAS} current controls the value of V_{GATE} applied to the power FET M_{CP1}, thus enabling selection of different increased values of R_{ON}.

[0065] In some embodiments, reduced gate-drive operation of a power FET in the ON state to limit current spikes during potentially damaging events may be enabled (triggered) by a control circuit (not shown) as a function of a measured parameter, such as the value of V_{IN}, V_{OUT}, pump capacitor voltages, or load current, and/or as the result of sensed events, such as short circuit events and/or charge imbalances on the pump capacitors. In some embodiments, reduced gate-drive operation of a power FET in the ON state to limit current spikes during potentially damaging events may be enabled (triggered) based on an external control signal for switch Sw_{RGD} that is asserted in advance of a known impending event, such as dynamic switching of conversion ratios.

[0066] The duration of reduced gate-drive operation for the power FETs may be set as a fixed time suitable for a particular application or may be determined based on some criteria. For example, reduced gate-drive operation for the power FETs may be a function of output load, or a function of output load and a selected maximum duration (i.e., a time-out parameter), or a function of the voltage across the fly capacitors having reached some percentage (e.g., 95%) of a desired target level, or some combination of these values and/or the values of other parameters.

[0067] An advantage of using diode-connected FETs in the voltage control circuit **310** fabricated using the same technology as the power FETs (e.g., NMOSFET) is that the devices essentially have matching characteristics with respect to process/voltage/temperature (PVT) variations.

[0068] In summary, the principal function of the gate driver circuit **306** is to enable at least two different voltage levels at Node A to be coupled to the gate of FET M_{LDO2}. More specifically, the voltage control circuit **310** can selectably shift the voltage at Node A between a first voltage level, in which the voltage control circuit **310** is not engaged (switch Sw_{RGD} is OPEN) and at least a second voltage level, in which the voltage control circuit **310** is engaged (switch Sw_{RGD} is CLOSED).

[0069] It should be appreciated that while the second LDO section **306** illustrated in FIG. **3** is preferred as simple to implement, requiring little power and circuit area, other devices or circuits that provide the same or similar function may be used in other embodiments. For example, Node A could be coupled through switch Sw_{RGD} to an amplifier having a level-shifted reference voltage as an input; the gate voltage to FET M_{LDO2} would be more accurate but at the expense of complexity, circuit area, and power (and thus efficiency).

[0070] Note that the LDO power supply **206** of FIG. **3**, including the second LDO section **306**, may be used to power all of the FET switches in a power converter **100** to limit currents through such switches as may be needed (for example, when dynamically changing the conversion ratio of the power converter **100**). In some instances, a level shifter **202** circuit would not be required for some FET switches (e.g., switches **S5** and **S7** in FIG. **1**), in which case φ may be applied directly to the associated pre-driver **204***a*.

Enhanced LDO Power Supply Embodiments with Reduced Gate-Drive Capability

[0071] In many applications, the LDO power supply **206** of FIG. **3** works perfectly well at a specified fixed switching frequency (e.g., 400 kHz). However, in some applications, it may not be desirable to switch at the specified fixed frequency during a startup or transition period. In light of the fact that the LDO power supply **206** is an open loop implementation, a possible operational limitation arises since the current limiting function (reduced gate-drive) of the circuit may vary depending on the switching frequency of the associated power converter **100**.

[0072] For example, FIG. **4**A is a graph **400** showing the gate voltage V_{GS} for the FET M_{CP1} and the current I_{MAIN} through FET M_{CP1} as a function of time at a switching frequency of 200 kHz in a reduced gate-drive mode of operation. Graph line **402** shows that the rise time of V_{GS} in the reduced gate-drive mode is about 300 nS (compared to about 5-10 nS in full gate-drive mode) at the selected frequency of operation for the associated power converter **100**. Graph line **404** shows that the corresponding limited current flow through FET M_{CP1} due to the reduced gate-drive voltage is fully settled at the selected switching frequency by the time Vas is settled. In the illustrated example, the resulting average current I_{AVG} at the selected frequency of operation is about 850 mA.

[0073] FIG. **4**B is a graph **410** showing the gate voltage V_{GS} for FET M_{CP1} and the current I_{MAIN} through FET M_{CP1} as a function of time at a switching frequency of 1 MHz in a reduced gate-drive mode of operation. Graph line **412** shows that the rise time of Vas in the reduced gate-drive mode is about 300 nS (note that the time scales in FIGS. **4**A and **4**B differ) at the selected frequency of operation for the associated power converter **100**. Graph line **414** shows the corresponding limited current flow through FET M_{CP1} due to the reduced gate-drive voltage still rising and not being settled at the higher switching frequency by the time Vas is settled. In the illustrated example, the resulting average current I_{AVG} at the selected frequency of operation is about 400 mA. This lower amount of average current I_{AVG} through FET M_{CP1} during a soft-start results in a slow power-up of the power converter **100**. Accordingly, the IC may not meet a customer's specification. In addition, if I_{AVG} is much lower than a typically-designed current value in some IC PVT corners and operating conditions, then the LDO power supply **206** may not output the minimum sourcing current required to support a load current.

[0074] The frequency dependency of the LDO power supply **206** may allow a large current spread between the minimum and maximum current allowed through FET M_{CP1}. Other possible contributors to a large current spread include mismatches between the devices (for example, a mismatch between M_{D0} and M_{LDO2} and/or between M_{DN} and M_{CP1}) which may also result in different current limits on different devices, and the dependency of the gate rising time for FET M_{CP1} on process and temperature variations, which adds extra spread to the device current limit. A large current spread may allow the current I_{MAIN} through FET M_{CP1} to exceed the current handling capability of the FET, or to fall below the minimum current required to support an output load. A large current spread also affects device reliability and may cause a drop in output voltage at full output load.

[0075] The present invention provides several enhanced LDO power supply circuits having a reduced gate-drive capability, along with corresponding methods, that accurately limit power FET current in a reduced gate-drive mode of operation and are independent of switching frequency as well as device mismatches and PVT variations. The inventive LDO power supply circuits provide reliable operation of power FETs within a power converter during dynamic re-configuration of the conversion ratio of the power converter, and also provide a level output voltage (essentially without drop) when a power converter operates at full load in different operating and PVT conditions.

[0076] In contrast to the open-loop implementation of the LDO power supply **206** of FIG. **3**, embodiments of the present invention utilize an added closed-loop feedback circuit and/or an added calibrated compensation circuit to closely control the control voltage V_{GATE} applied to the gate of the power FET M_{CP1}. Depending on the nature of the added circuit, connection to the LDO power supply **206** may be made at one of several nodes. For example, FIG. **5** is a block diagram of an LDO power supply circuit **500** showing possible connections nodes A, B, and C for added feedback or compensation circuits.

[0077] Connecting a feedback or compensation circuit at node A allows the gate voltage V_{GS_SF} of the source-follower FET M_{LDO2} to be regulated to control V_{LDO_OUT2} to the final inverter **204***b* so that V_{GATE} is adjusted to provide a reduced gate-drive to the power FET M_{CP1} that is substantially independent of frequency. Connecting a feedback or compensation circuit at node B allows V_{LDO_OUT2} to the final inverter **204***b* to be directly regulated so that V_{GATE} is adjusted to provide a reduced gate-drive to the power FET M_{CP1} that is substantially independent of frequency. Connecting a feedback or compensation circuit at node C allows V_{GATE} to be directly set to provide a reduced gate-drive to the power FET M_{CP1} that is substantially independent of frequency. Details of several example embodiments of such feedback or compensation circuits are set forth in the following subsections.

A. Closed-Loop Direct Supplementation of V_{GS} SF

[0078] FIG. **6**A is a block diagram of a first embodiment of an enhanced LDO power supply **600** utilizing a closed-loop direct supplementation of V_{GS_SF}. Similar in most aspects to the LDO power supply **206** of FIG. **3**, the enhanced LDO power supply **600** differs by including a closed-loop supplemental variable current source **602** coupled to Node A, configured to directly provide an offset current I_{OFFSET} designed to increase or decrease the gate voltage V_{GS} SE of the source-follower FET M_{LDO2}. The offset current I_{OFFSET} is proportional to a target reference current I_{TARGET} minus either I_{MAIN} or a current value that is proportional to I_{MAIN}. Thus, $I_{OFFSET}=K^*(I_{TARGET}-I_{MAIN})$, where K is a proportionality constant that may represent, for example, the size ratio of FET M_{DN} (inside the voltage control circuit **310**) to FET M_{CP1}.

[0079] In operation, the second LDO section **306** is initially biased by I_{BIAS}. The current through power FET M_{CP1} is compared with a target reference current and the feedback offset current I_{OFFSET} is generated based on any difference. In the reduced gate-drive mode, any generated value of I_{OFFSET} adds to I_{BIAS} and, when applied through the voltage control circuit **310**, increases or decreases the gate voltage V_{GS_SF} of the source-follower FET M_{LDO2}, thereby increasing I_{MAIN} until I_{MAIN} equates to I_{TARGET}. Accordingly, FIG. **6**A is an example of a closed-loop direct supplementation of V_{GS_SF}.

[0080] The feedback loop adapts I_{MAIN} regardless of frequency. As an example, FIG. **6**B is a graph **620** showing the current I_{MAIN} through FET M_{CP1} as a function of time at a switching frequency of 1 MHz in a reduced gate-drive mode of operation. In this example, the average current I_{AVG} at the selected frequency of operation is about 400 mA (dotted line **622**), but ramps up to about 850 mA (dotted line **624**) when the feedback offset current I_{OFFSET} is generated and applied (compare FIG. **4**A, showing I_{AVG} equals about 850 mA at 200 KHz). The supplemental feedback offset current I_{OFFSET} increases the gate voltage Vas SE of the source-follower FET M_{LDO2}, thereby increasing I_{MAIN} for a time.

[0081] A number of circuits may be used for the closed-loop supplemental variable current source **602**. For example, FIG. **7**A is a block diagram of an example circuit for generating the supplemental feedback offset current I_{OFFSET}. The current through the power FET M_{CP1}, I_{MAIN}, is coupled to a current-to-voltage converter **702** which generates an output voltage V_{CP1} that is proportional to the magnitude of I_{MAIN}. After filtering through a low-pass filter **704** to obtain an average value, V_{CP1} is coupled to a first input of an operational transconductance amplifier (OTA) **706**. A second input to the OTA **706** is a reference voltage VREF that is proportional to a target reference current I_{TARGET}. The OTA **706** sources or sinks an output current I_{OFFSET} whose magnitude is proportional to the differential input voltages. The output current I_{OFFSET} is applied to Node A in the circuit of FIG. **6**A as described above.

[0082] FIG. **7**B is a schematic diagram of a first example embodiment of a current-to-voltage converter **702**. The output of the final inverter **204***b*, V_{GATE}, is shown coupled to the gate of power FET M_{CP1} and to the gate of a scaled replica FET, M_{REP}. The scaled replica FET M_{REP} may be, for example, 0.001% the size of power FET M_{CP1}, and accordingly, has little effect on the operation of the power FET M_{CP1} but does conduct a current, I_{SENSE}, that is proportional to I_{MAIN}. Applying I_{SENSE} through a resistor R generates the output voltage V_{CP1}.

[0083] FIG. **7**C is a schematic diagram of a second example embodiment of a current-to-voltage converter **702**. The output of the final inverter **204***b*, V_{GATE}, is shown coupled to the gate of power FET M_{CP1} and to the gate of a scaled replica FET, M_{REP}. The scaled replica FET M_{REP} may be, for example, 0.001% the size of power FET M_{CP1}, and conducts a current, I_{SENSE}, that is proportional to I_{MAIN}. Both I_{SENSE} and I_{MAIN} are coupled to respective first and second inputs of a differential amplifier **708** that generates an output voltage V_{OUT} proportional to the difference between I_{SENSE} and I_{MAIN}. Output voltage V_{OUT} is coupled to the gate of a feedback FET, M_{FB}, the source of which is coupled to the first input of the differential amplifier **708**. The drain of feedback FET M_{FB} is coupled to a resistor R, thus generating the output voltage V_{CP1}. An advantage of the current-to-voltage converter **702** shown in FIG. **7**C is that it maintains the same drain-to-source voltage V_{DS} across the power FET M_{CP1} and scaled replica FET M_{REP} and thus generates a more accurate sense current I_{SENSE}.

B. Closed-Loop Indirect Supplementation of V_{GS} SF

[0084] FIG. **8** is a block diagram of a second embodiment of an enhanced LDO power supply **800**. Similar in many aspects to the LDO power supply **206** of FIG. **3**, the enhanced LDO power supply **600** differs by including circuitry for generating a closed-loop indirect supplementation of V_{GS_SF}.

[0085] In greater detail, separate full gate-drive and reduced gate-drive current sources are provided. More specifically, a full gate-drive current source I_{FGD} coupled in series with a diode D between V_{DD} and a floating reference potential **302**, such as circuit ground, drives the gate of FET M_{LDO2} when full gate-drive switch Sw_{FGD} is closed and reduced gated-drive switch Sw_{RGD} is open. Conversely, in the reduced gate-drive mode, full gate-drive switch Sw_{FGD} is open and reduced gated-drive switch Sw_{RGD} is closed.

[0086] For the reduced gated-drive mode, a current source I_{BIAS} is coupled in series with a resistor R**1** between V_{DD} and the floating reference potential **302**. The voltage generated across the resistor R**1** is coupled to a first input of a differential amplifier **802**. The output of the differential amplifier **802** is selectably couplable to Node A, and thus to the gate of the FET M_{LDO2}, through the reduced gated-drive switch Sw_{RGD}.

[0087] A second input of the differential amplifier **802** is coupled to a feedback circuit **804** that includes a variable current source I_{SENSE} coupled in series with an RC circuit between V_{DD} and the floating reference potential **302**. The RC circuit, comprising a parallel resistor R**2** and capacitor C**2**, acts as a filter and provides an average voltage at the second input of the differential amplifier **802**. The current source I_{SENSE} may be, for example, a scaled replica FET, M_{REP} having a gate coupled to the output V_{GATE} of the final inverter **204***b*, as in FIG. **7**B. Thus, I_{SENSE} is proportional to I_{MAIN} (e.g., $1/10{,}000^{th}$ of I_{MAIN}). The I_{SENSE} current applied to resistor R**2** and capacitor C**2** provides an averaged voltage input to the differential amplifier **802**. The differential amplifier **802** corrects the difference between I_{BIAS} and the closed-loop feedback current I_{SENSE} by setting up a corresponding V_{GS_SF} voltage. Note that in the embodiment of FIG. **8**, the Zener diode D**2** of the embodiments shown in some of the other novel embodiments is not required in reduced gated-drive mode, since the output of the differential amplifier **802** is set to a desired level to provide a reduced gate drive voltage. An advantage of the embodiment of FIG. **8** is that it generally has a low output impedance and high feedback loop stability.

C. Example Compensation Circuit Coupled to Node A

[0088] Another approach to providing a reduced gate-drive to a power FET M_{CP1} that is substantially independent of frequency is to apply a calibrated compensation voltage boost to briefly increase the drive capability of an LDO power supply **206** to offset the slow rise time of V_{GATE} from the final inverter **204***b* caused by the relatively large gate capacitance of the power FET M_{CP1} (e.g., about 20 μF in some MOSFET implementations). State another way, the compensation voltage boost briefly increases the reduced gate-drive voltage to the gate of the power FET to compensate for the gate capacitance of the power FET.

[0089] For example, FIG. **9**A is a graph of V_{GATE} as a function of time without the benefit of a compensation voltage boost. As arrow A indicates, the need to charge the relatively large gate capacitance of the power FET M_{CP1} leads to a relatively slow rise in V_{GATE}. FIG. **9**B is a graph of Vas SE as a function of time, showing the periodic augmentation to the normal $2V_{GS}$ voltage that a compensation voltage boost V_{CHG} provides if applied at Node A. FIG. **9**C is a graph of V_{LDO_OUT2} as a function of time, showing the periodic augmentation in the normal reduced gate-drive output voltage resulting from application of an appropriately scaled compensation voltage boost V_{CHG} to Node A or to Node B (see FIG. **5**). FIG. **9**D is a graph of V_{GATE} as a function of time with the benefit of a compensation voltage boost to V_{LDO_OUT2}. As arrow B indicates, the added compensation voltage boost V_{CHG} compensates for the otherwise relatively slow rise in V_{GATE}, as shown in FIG. **9**A, such that V_{GATE} has a fast, sharp rise. After the compensation voltage boost V_{CHG} times out, the reduced gate-drive to the power FET M_{CP1} from the final inverter **204***b*, as powered by V_{LDO_OUT2} from the second LDO section **306**, continues for the rest of a cycle.

[0090] FIG. **10** is a block diagram of a third embodiment of an enhanced LDO power supply **1000**. Similar in most aspects to the LDO power supply **206** of FIG. **3**, the enhanced LDO power supply **1000** differs by including a calibrated compensation circuit **1002** coupled to Node A and configured to periodically boost the output voltage V_{LDO_OUT2} to the final inverter **204***b*. The calibrated compensation circuit **1002** includes a generally fixed current source I_{CHG} coupled in series with an RC circuit (comprising a parallel resistor R**2** and capacitor C**2** in this example) between V_{DD} and the floating reference potential **302**. A switch Sw_{BOOST} selectively couples the voltage across the resistor R**2** and capacitor C**2** to Node A.

[0091] In operation, the capacitor C**2** is pre-charged to a level above the $2V_{GS}$ voltage provided by the voltage control circuit **310**. The switch Sw_{BOOST} is periodically closed for a short time when V_{GATE} is rising (e.g., the state of switch Sw_{BOOST} may be set or triggered by a control signal that tracks the state of the input φ to the level shifter, if present, and to the driver circuit **204**). Closing switch Sw_{BOOST} transfers charge from the capacitor C**2** to capacitor C_{VGS}, and thus adds an additional voltage, V_{CHG}, to Vas SF. The additional voltage V_{CHG} applied to the gate of the source-follower FET M_{LDO2} compensates (adds to) the charge that is transferred from the LDO capacitor C_O to the gate capacitance of the power FET M_{CP1} very quickly when the power FET is ON. This charge transference makes the edges of V_{GATE} sharper as shown in FIG. **9**D and results in the current I_{MAIN} through the power FET M_{CP1} being substantially independent of frequency in the reduced gate-drive mode.

[0092] The amount of charge stored in capacitor C**2** and the duration of the closed state for the switch Sw_{BOOST} needed for a particular application may be fairly precisely determined by modeling the enhanced LDO power supply **1000**, and in that sense the circuit is calibrated.

D. Example Compensation Circuit Coupled to Node B

[0093] FIG. **11** is a block diagram of a fourth embodiment of an enhanced LDO power supply **1100**. Similar in most aspects to the enhanced LDO power supply **1000** of FIG. **10**, the enhanced LDO power supply **1100** differs by coupling the calibrated compensation circuit **1002** at Node B (which periodically boosts V_{LDO_OUT2}) rather than at Node A (which periodically boosts the gate voltage V_{GS_SF} to the source-follower FET M_{LDO2}). Connecting the calibrated compensation circuit **1002** at node B allows V_{LDO_OUT2} to the final inverter **204***b* to be directly regulated so that V_{GATE} is adjusted to provide a reduced gate-drive to the power FET M_{CP1} that is substantially independent of frequency. Again, the amount of charge stored in capacitor C**2** and the duration of the closed state for the switch Sw_{BOOST} needed for a particular application may be fairly precisely determined by modeling the enhanced LDO power supply **1000**, and in that sense the circuit is calibrated.

[0094] In a variant embodiment, capacitor C**2** may be omitted, since the current source I_{CHG} can periodically add charge through the switch Sw_{BOOST} to the capacitor C_O to compensate (add to) the charge that is transferred from the LDO capacitor C_O to the gate capacitance of the power FET M_{CP1}.

E. Example Compensation Circuit Coupled to Node C

[0095] FIG. **12** is a block diagram of a fifth embodiment of an enhanced LDO power supply **1200**. In essence, during the normal full gate-drive mode, the first LDO section **300** is enabled to provide V_{GATE} to the power FET M_{CP1} while the second LDO section **306** is disabled. Conversely, during the reduced gate-drive mode, the second LDO section **306** is enabled to provide V_{GATE} to the power FET M_{CP1} while the first LDO section **300** is disabled.

[0096] In the illustrated example, the first LDO section **300** provides V_{LDO_OUT1} to the level shifter **202**, the pre-driver **204***a*, and the final driver **204***b*, and the output of the second LDO section **306** is coupled to Node C (i.e., to the gate of the power FET M_{CP1}). The final driver **204***b* is shown modified to provide the clock signal φ to the PMOS FET p in the final driver **204***b* through an OR gate **1102** only if a control signal, EN_{RGD}, is asserted with a LOW (logical 0) value.

[0097] The second LDO section **306** is configured similar to the enhanced LDO power supply **1000** of FIG. **10**. However, the calibrated compensation circuit **1002** of FIG. **10** is omitted, the source of the source-follower FET M_{LDO2} is connected to Node C, and power to the source-follower FET M_{LDO2} is controlled by a PFET **M1** coupled between V_{DD} and the drain of the source-follower FET M_{LDO2}. The ON or OFF state of FET **M1** is controlled by an OR gate **1104** to which is coupled a control signal $\overline{EN_{RGD}}$ (the complement of the EN_{RGD} control signal to the OR gate **1102**) and the output of the pre-driver **204***a*. Further, the voltage control circuit **310** is modified to omit the switch Sw_{RGD} of FIG. **10** (not needed since the entire second LDO section **306** is only operational during the reduced gate-drive mode) and the diode D**2** and capacitor C_O shown in FIG. **10** are omitted (not needed since the entire second LDO section **306** is only operational during the reduced gate-drive mode and there is a direct connection to Node C). Lastly, an optional switch SW_{RGD2} is interposed between Noda A and the current source I_{BIAS}, and may be opened to disable the voltage control circuit **310** (alternatively, the PFET **M1** may be disabled using the $\overline{EN_{RGD}}$ control signal, allowing the switch SW_{RGD2} to be omitted).

[0098] In full gate-drive mode, the control signal EN_{RGD} has a LOW (logical 0) value. Accordingly, PFET **M1** is set to an OFF state by the complementary control signal $\overline{EN_{RGD}}$ and coupling of the clock signal φ through the OR gate **1102** to the PMOS FET p in the final driver **204***b* is enabled. In addition, the switch SW_{RGD2} is opened. Thus, the first LDO section **300** is enabled and the second LDO section **306** is disabled.

[0099] In reduced gate-drive mode, the control signal EN_{RGD} has a HIGH (logical 1) value. Accordingly, PFET **M1** is set to an ON state by the complementary control signal $\overline{EN_{RGD}}$ and coupling of the clock signal φ through the OR gate **1104**, PFET **M1**, and source-follower FET M_{LDO2} to the gate of the power FET M_{CP1} is enabled. In addition, the switch SW_{RGD2} is closed. Thus, the first LDO section **300** is disabled and the second LDO section **306** is enabled. The value of V_{GS_SF} is set by the voltage control circuit **310** to $2V_{GS}$ and sets V_{GATE} when the power FET M_{CP1} is ON to a targeted reduced gate-drive level.

[0100] The embodiment of FIG. **12** has an advantage that it provides an additional boost to V_{GS}_SF through PFET **M1** and the gate-drain capacitance C_{GD} of M_{LDO2} without extra circuitry while the power FET M_{CP1} turns ON. When PFET **M1** is ON, then the drain of M_{LDO2} is pulled up to V_{DD}. The C_{GD} capacitance of M_{LDO2} couples the rise in charge at the drain of M_{LDO2} to V_{GS_SF} for a short duration, which provides additional charge to the gate of the power FET M_{CP1}.

Alternative Embodiments

[0101] It should be appreciated that both feedback and compensation circuits may be used together. For example, in some applications, it may be useful to combine the closed-loop direct supplementation of V_{GS_SF} as shown in FIG. **6**A or FIG. **8** with the compensation circuit **1002** shown in FIG. **10** (Node A connected) or FIG. **11** (Node B connected).

[0102] Further, while the embodiments of this disclosure have focused on the use of a source-follower FET M_{LDO2}, alternative embodiments may use any circuit (e.g., an op amp, OTA, etc.) that provides a reduced gate drive voltage to the final inverter **204***b* or to the gate of the power FET M_{CP1} in response to a change in a reference voltage.

[0103] Embodiments of the invention may also be used to accurately limit startup currents in buck/boost power converters and LDO power supplies in general. For example, FIG. **13** is a schematic diagram of an LDO **1300**. A current source I_{REF} is coupled in series with an RC circuit **1302** (comprising a parallel resistor Ra and capacitor Ca) between V_{DD} and a floating reference potential. A variable current source I_{OFFSET} is coupled between V_{DD} and the RC circuit **1302**. The current source I_{OFFSET} may be implemented as described in FIGS. **6**A and **7**A-**7**C. The current from I_{REF} and I_{OFFSET} when applied across the RC circuit **1302** generates a voltage VREF, which is coupled to a first input of a differential amplifier **1304**. The output of the differential amplifier **1304** is coupled to the gate of a power FET M. The conduction channel of FET M is coupled in series with resistors Rb and Rc between V_{IN} and the floating reference potential. A feedback voltage from between resistors Rb and Rc is coupled to a second input of the differential amplifier **1304**. In operation, the current source I_{OFFSET} adjusts the current across the RC circuit **1302** that generates VREF as a function of the difference between I_{TARGET} and I_{MAIN} until the difference is essentially eliminated.

Benefits

[0104] Embodiments of the present invention may some or all of the following advantages: a current limiting value that is independent of a power converter switching frequency, device mismatches and process, voltage, and/or temperature (PVT) variations; accurate soft-start current limiting; reliable operation of power FETs in power converters having reconfigurable conversion ratios when dynamically changing conversion ratios; and the ability to keep a level output voltage when a power converter operates at full load in different operating and PVT conditions.

Circuit Embodiments

[0105] Circuits and devices in accordance with the present invention may be used alone or in combination with other components, circuits, and devices. Embodiments of the present invention may be fabricated as integrated circuits (ICs), which may be encased in IC packages and/or in modules for case of handling, manufacture, and/or improved performance. In particular, IC embodiments of this invention are often used in modules in which one or more of such ICs are combined with other circuit components or blocks (e.g., filters, amplifiers, passive components, and possibly additional ICs) into one package. The ICs and/or modules are then typically combined with other components, often on a printed circuit board, to form part of an end-product such as a cellular telephone, laptop computer, or electronic tablet, or to form a higher-level module which may be used in a wide variety of products, such as vehicles, test equipment, medical devices, etc. Through various configurations of modules and assemblies, such ICs typically enable a mode of communication, often wireless communication.

[0106] As one example of further integration of embodiments of the present invention with other components, FIG. **14** is a top plan view of a substrate **1400** that may be, for example, a printed circuit board or chip module substrate (e.g., a thin-film tile). In the illustrated example, the substrate **1400** includes multiple ICs **1402***a*-**1402***d* having terminal pads **1404** which would be interconnected by conductive vias and/or traces on and/or within the substrate **1400** or on the opposite (back) surface of the substrate **1400** (to avoid clutter, the surface conductive traces are not shown and not all terminal pads are labelled). The ICs **1402***a*-**1402***d* may embody, for example, signal switches, active filters, amplifiers (including one or more LNAs), and other circuitry. For example, IC **1402***b* may incorporate one or more instances of an enhanced LDO power supply circuit like the circuits shown in FIGS. **6**A, **8**, **10**, **11**, and/or **12**.

[0107] The substrate **1400** may also include one or more passive devices **1406** embedded in, formed on, and/or affixed to the substrate **1400**. While shown as generic rectangles, the passive devices **1406** may be, for example, filters, capacitors, inductors, transmission lines, resistors, planar antennae elements, transducers (including, for example, MEMS-based transducers, such as accelerometers, gyroscopes, microphones, pressure sensors, etc.), batteries, etc., interconnected by conductive traces on or in the substrate **1400** to other passive devices **1406** and/or the individual ICs **1402***a*-**1402***d*. The front or back surface of the substrate **1400** may be used as a location for the formation of other structures.

System Aspects

[0108] Embodiments of the present invention are useful in a wide variety of applications, including portable computing devices (e.g., laptops, notebooks, cell phones, tablets), datacenters and telecom centers that have battery backup systems, household appliances and electronics, vehicles (e.g., automobiles, drones, planes, boats, trains, ships), general purpose DC/DC and AC/DC power converters, and radio frequency (RF) circuits and systems.

[0109] Radio system usage may include wireless RF systems (including base stations, relay stations, and hand-held transceivers) that use various technologies and protocols, including various types of orthogonal frequency-division multiplexing ("OFDM"), quadrature amplitude modulation ("QAM"), Code-Division Multiple Access ("CDMA"), Time-Division Multiple Access ("TDMA"), Wide Band Code Division Multiple Access ("W-CDMA"), Global System for Mobile Communications ("GSM"), Long Term Evolution ("LTE"), 5G, 6G, and WiFi (e.g., 802.11a, b, g, ac, ax, be) protocols, as well as other radio communication standards and protocols.

Methods

[0110] Another aspect of the invention includes methods for protecting a power converter. For example, FIG. **15** is a process flow chart **1500** showing a first method for protecting a power converter. The method includes: controlling the ON resistance, R_{ON}, of a power FET in the power converter to lower the R_{ON} of the power FET in a first ON state during normal power converter operation and to raise the R_{ON} of the power FET in a second ON state to restrict flow of current through the power FET (Block **1502**); generating a feedback current proportional to a current flow through the power FET (Block **1504**); and regulating the R_{ON} of the power FET in the second ON state in proportion to the generated feedback current to provide a reduced gate-drive to the power FET that is substantially independent of frequency (Block **1506**).

[0111] Additional aspects of the above method may include one or more of the following: regulating the RON of the power FET in the second ON state in proportion to the generated feedback current during a dynamic re-configuration of a conversion ratio of the power converter; regulating the RON of the power FET in the second ON state in proportion to the generated feedback current during a startup period of the power converter; and/or regulating the RON of the power FET in the second ON state in proportion to the generated feedback current during a charge re-balancing event among two or more capacitors within the power converter.

[0112] As another example, FIG. **16** is a process flow chart **1600** showing a second method for protecting a power converter. The method includes: controlling the ON resistance, R_{ON}, of a power FET in the power converter to lower the R_{ON} of the power FET in a first ON state during normal power converter operation and to raise the R_{ON} of the power FET in a second ON state to restrict flow of current through the power FET (Block **1602**); and applying a compensation voltage boost to a gate of the power FET in the second ON state to increase a drive voltage to the gate to compensate for a gate capacitance of the power FET (Block **1604**).

[0113] Additional aspects of the above method may include one or more of the following: applying the compensation voltage boost during a dynamic re-configuration of a conversion ratio of the power converter; applying the compensation voltage boost during a startup period of the power converter; and/or applying the compensation voltage boost during a charge re-balancing event among two or more capacitors within the power converter.

[0114] As yet another example, FIG. **17** is a process flow chart **1700** showing a third method for protecting a power converter. The method includes: generating a feedback current proportional to a current flow through a power FET of the power converter when the power FET is in an ON state (Block **1702**); generating a reduced gate-drive voltage in proportion to the generated feedback current, wherein the reduced gate-drive voltage is substantially independent of frequency (Block **1704**); and coupling the reduced gate-drive voltage to a gate of the power FET to restrict flow of current through the power FET (Block **1706**).

[0115] Additional aspects of the above method may include one or more of the following: generating the reduced gate-drive voltage during a dynamic re-configuration of a conversion ratio of the power converter; generating the reduced gate-drive voltage during a startup period of the power converter; generating the reduced gate-drive voltage during a charge re-balancing event among two or more capacitors within the power converter; wherein the reduced gate-drive voltage increases an ON resistance, R_{ON}, of the power FET; and/or wherein the reduced gate-drive voltage causes the power FET to function as a controlled current-limited source when the power FET is in a saturation mode of operation.

Fabrication Technologies & Options

[0116] The term "MOSFET", as used in this disclosure, includes any field effect transistor (FET) having an insulated gate whose voltage determines the conductivity of the transistor, and encompasses insulated gates having a metal or metal-like, insulator, and/or semiconductor structure. The terms "metal" or "metal-like" include at least one electrically conductive material (such as aluminum, copper, or other metal, or highly doped polysilicon, graphene, or other electrical conductor), "insulator" includes at least one insulating material (such as silicon oxide or other dielectric material), and "semiconductor" includes at least one semiconductor material.

[0117] As used in this disclosure, the term "radio frequency" (RF) refers to a rate of oscillation in the range of about 3 kHz to about 300 GHz. This term also includes the frequencies used in wireless communication systems. An RF frequency may be the frequency of an electromagnetic wave or of an alternating voltage or current in a circuit.

[0118] With respect to the figures referenced in this disclosure, the dimensions for the various elements are not to scale; some dimensions may be greatly exaggerated vertically and/or horizontally for clarity or emphasis. In addition, references to orientations and directions (e.g., "top", "bottom", "above", "below", "lateral", "vertical", "horizontal", etc.) are relative to the example drawings, and not necessarily absolute orientations or directions.

[0119] Various embodiments of the invention can be implemented to meet a wide variety of specifications. Unless otherwise noted above, selection of suitable component values is a matter of design choice. Various embodiments of the invention may be implemented in any suitable integrated circuit (IC) technology (including but not limited to MOSFET structures), or in hybrid or discrete circuit forms. Integrated circuit embodiments may be fabricated using any suitable substrates and processes, including but not limited to standard bulk silicon, high-resistivity bulk CMOS, silicon-on-insulator (SOI), and silicon-on-sapphire (SOS). Unless otherwise noted above, embodiments of the invention may be implemented in other transistor technologies such as bipolar, BiCMOS, LDMOS, BCD, GaAs HBT, GaN HEMT, GaAs pHEMT, and MESFET technologies. However, embodiments of the invention are particularly useful when fabricated using an SOI or SOS based process, or when fabricated with processes having similar characteristics. Fabrication in CMOS using SOI or SOS processes enables circuits with low power consumption, the ability to withstand high power signals during operation due to FET stacking, good linearity, and high frequency operation (i.e., radio frequencies up to and exceeding 300 GHz). Monolithic IC implementation is particularly useful since parasitic capacitances generally can be kept low (or at a minimum, kept uniform across all units, permitting them to be compensated) by careful design.

[0120] Voltage levels may be adjusted, and/or voltage and/or logic signal polarities reversed, depending on a particular specification and/or implementing technology (e.g., NMOS, PMOS, or CMOS, and enhancement mode or depletion mode transistor devices). Component voltage, current, and power handling capabilities may be adapted as needed, for example, by adjusting device sizes, serially "stacking" components (particularly FETs) to withstand greater voltages, and/or using multiple components in parallel to handle greater currents. Additional circuit components may be added to enhance the capabilities of the disclosed circuits and/or to provide additional functionality without significantly altering the functionality of the disclosed circuits.

CONCLUSION

[0121] A number of embodiments of the invention have been described. It is to be understood that various modifications may be made without departing from the spirit and scope of the invention. For example, some of the steps described above may be order independent, and thus can be performed in an order different from that described. Further, some of the steps described above may be optional. Various activities described with respect to the methods identified above can be executed in repetitive, serial, and/or parallel fashion.

[0122] It is to be understood that the foregoing description is intended to illustrate and not to limit the scope of the invention, which is defined by the scope of the following claims, and that other embodiments are within the scope of the claims. In particular, the scope of the invention includes any and all feasible combinations of one or more of the processes, machines, manufactures, or compositions of matter set forth in the claims below. (Note that the parenthetical labels for claim elements are for ease of referring to such elements, and do not in themselves indicate a particular required ordering or enumeration of elements; further, such labels may be reused in dependent claims as references to additional elements without being regarded as starting a conflicting labeling sequence).

What is claimed is:

1. A gate control circuit for regulating the ON resistance, R_{ON}, of a power FET, including:

(a) a source-follower circuit including a source-follower FET having a conduction channel configured to be coupled to the gate of the power FET and configured to selectively apply at least a first voltage or a second voltage to the gate of the power FET such that the R_{ON} of the power FET in an ON state is lower when the first voltage is applied and higher when the second voltage is applied; and

(b) a compensation circuit coupled to one of a gate of the source-follower FET or an output of the source-follower FET and configured to apply a compensation voltage boost to adjust the second voltage to the gate of the power FET to compensate for a gate capacitance of the power FET.

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